

CMO Primer

An update for the age of the floater

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- Since our last update, the CMO market has come to be dominated by floater issuance; we expand our primer to cover the structures created as by-products of these floaters or as a result of this floater dominance, namely inverse floaters, mountains, TTIBs, and IRCs
- We also add a section on GMCs given their recent resurrection; almost \$11.5bn were issued in the first 6 months of 2026
- We will separately post an excel spreadsheet to J.P. Morgan Markets that shows the cashflows for different structures with flexible assumptions around prepayment speeds and tranche sizing, with some of these new structures added
- This primer walks through the process of creating a CMO and the details of the structures (stripdowns, SEQs, PACs, TACs, LCFs, VADM, FLTs and their companion principal-bearing structures, supports, etc.)
- We also cover CMO derivatives (IOs & IIOs), as well as the mechanics of excess IO transactions
- Also included is an appendix on the tools we have for analyzing CMOs and tracking historical performance; J.P. Morgan's DataQuery offers historical pricing spreads on principal bearing classes and analytics on our CMO, IO, and IIO indices
- For more regular updates on the market, please see our monthly CMO Issuance Report and our MSR & IO Monthly Chartbook

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See page 45 for analyst certification and important disclosures.

CMO Primer

Since our last update, floater issuance has become a driving force of CMO production. In this writeup, we expand our primer to cover the structures created as by-products of these floaters or as a result of this floater dominance, namely inverse floaters, mountains, TTIBs, and IRCs. Although many of these products existed before this period, they were not issued in any meaningful size. Given their increased issuance in today's CMO market, we think these structures are worth exploring in depth. We also add a section on GMCs, another product recently resurrected with a substantial amount of issuance in 2026.

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I. Putting structure in context: A brief update on the CMO market

Issuance

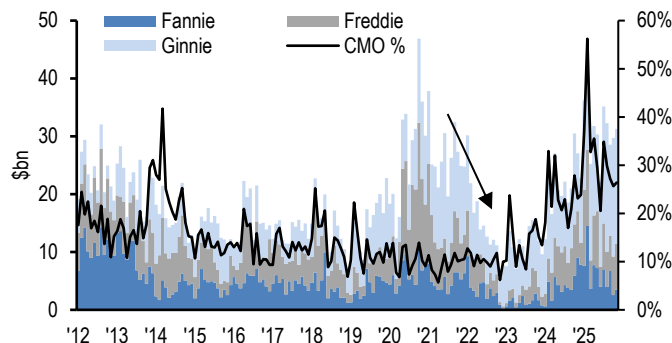
Historically, principal-bearing CMO gross issuance has made up 10–15% of total fixed-rate agency MBS gross issuance. From 2023 onward, however, principal-bearing CMOs have accounted for 20–25% of monthly issuance, driven by large volumes in 2025 and 2026 (**Figure 1**). This higher share has been propelled by floater issuance, which reflects banks' need for short-duration assets (**Figure 2**). Floater issuance has skewed toward Ginnie Mae, owing to continued favorable regulatory treatment of Ginnie securities for banks, though conventional issuance has also been sizeable over this

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period.

Figure 1: Annual issuance has varied throughout the market's history, but from 2023 onwards, CMOs have absorbed roughly 20-25% of the available supply of agency mortgage pass-throughs ...

CMO issuance by agency (\$bn, left) compared to the overall share of fixed-rate MBS PTs issued (% , right)

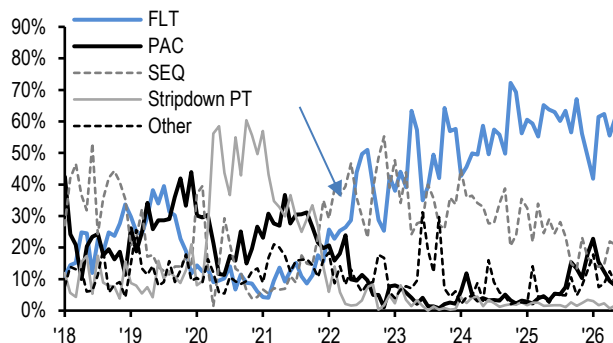


Source: J.P. Morgan, Fannie Mae, Freddie Mac, Ginnie Mae

Figure 2: ... and the increase in CMO issuance over the past several years has been driven by floaters

The increase in CMO issuance over the past several years has been driven by floaters

Issuance broken down by CMO type. For classification purposes, we use the waterfall: Floater, PAC, Seq, Stripdown, and bucket the rest into "Other"



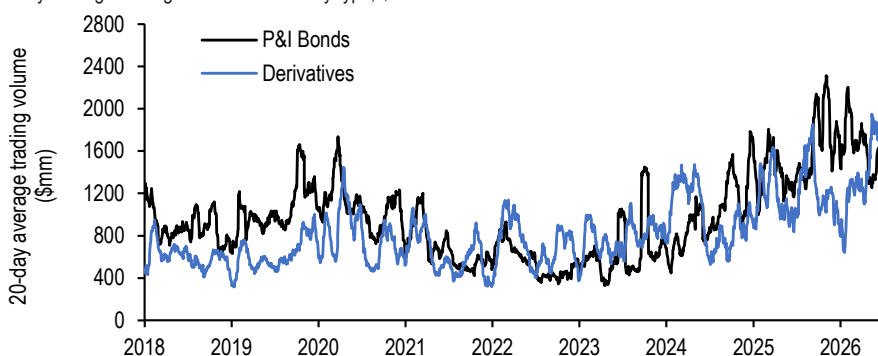
Source: J.P. Morgan, Fannie Mae, Freddie Mac, Ginnie Mae

Liquidity

Aside from the robust primary market in CMOs, a substantial amount of trading takes place on the secondary side. TRACE provides trading volumes in CMO by agency and by principal-bearing bonds versus derivatives. **Figure 3** shows that the monthly average (in notional terms) has lately been roughly \$1.2bn a day for principal & interest bonds and over \$1.3bn for derivatives.

Figure 3: Agency CMO trading volumes

20-day average trading volume in CMOs by type, \$mm



Source: J.P. Morgan, TRACE

II. An Overview of the CMO Market

Although mortgage pass-throughs have become one of the most successful products in U.S. fixed income, one size does not fit all. For some, the average life of a new pass-through is too long. For others, too short. For some, the prepayment risk is too great. Others would welcome even greater exposure. Some want more yield than the most readily available pass-throughs provide. Others have sensitivity to the dollar price of the asset.

Collateralized Mortgage Obligations (CMOs) allow investors to customize both the risks and rewards of owning mortgages. A CMO takes the principal and interest from a pool of assets—mortgage loans, mortgage pass-throughs, or even other CMOs—and allocates those cash flows to different CMO classes or tranches. Every CMO reduces to a set of rules, detailed in the CMO prospectus, for allocating the cash flows of the underlying collateral. Some of the resulting classes have more duration than the underlying asset, and others have less. Some have more negative convexity, and others less. Some bear greater prepayment risk, and other less. CMOs can differ dramatically in characteristics like average life, yield, coupon, and price. The structured MBS market takes the risks of the available pass-through or other collateral, breaks it apart and distributes it to investors most motivated to bear the component parts.

Because CMOs can tailor mortgage cash flows to the needs of a wide range of investors, the market has flourished since its inception in 1984. CMOs have become important in constructing the portfolios of banks, insurance companies, pension funds, money managers, and other investors both in the U.S. and worldwide. Annual issuance has varied throughout the market's history, but from 2023 onwards, CMOs have absorbed roughly 20-25% of the available supply of agency mortgage pass-throughs.

With an agency CMO, the principal and interest are guaranteed by either Fannie Mae, Freddie Mac, or Ginnie Mae. Just as with a pass-through, the guarantee relieves the investor of any need to master the credit risk of the asset. Each tranche is also tradable as an individual security with its own CUSIP, maturity, coupon, and other traditional features.

III. The Process of Creating a CMO

Because of the range of possible structures and the ready availability of pass-throughs, few markets match agency CMOs in customizing cash flows to individual portfolio needs. In corporate bonds, the issuer often determines the timing of a bond issuance and most of its features. Most features of an agency CMO—timing, collateral, and structure—can change to meet investor needs, although change can incur costs in the form of price, yield, liquidity, or their combinations.

A CMO usually comes out of an ongoing dialogue between structurers and a range of current and potential investors. Investors reflect their changing preferences to CMO structuring desks to see about the possibility of creating a new class tailored to the investor's needs. A structurer also needs to estimate where it can sell the cash flows left over after creating a class to meet a particular investor's needs, helped along by the presence of secondary markets for these securities. After all, customizing the features of one class can affect a range of other classes. If customizing one class impairs the pricing of others, then the structuring desk will need to charge the inquiring investor to compensate for the possible losses on other classes.

A structuring desk also keeps a careful eye of the price of pass-throughs, the raw material for creating a CMO. If the price of a pass-through rises or falls without any change in the market price of certain CMO classes, then the economics of creating a new class may change dramatically. A structuring desk may respond to inquiry by coming back with a price on a bond built to an investor's exact specifications, or by suggesting alternative specifications that might meet the investors needs more efficiently or at a lower price under prevailing market conditions.

In terms of pricing convention, fixed-rate, principal-bearing CMOs are typically quoted at a spread to the interpolated Treasury curve (I-spread on Bloomberg) based on the expected average life of the bond. That average life is calculated using the Bloomberg Median prepayment projection from dealer submissions for the underlying collateral, viewable through the VALL function. Prepayments are often quoted in “PSA” terms. 100 PSA is a prepayment vector that starts at 0.2c and increases by 0.2c each month until leveling out at 6c in month 30. Other speed profiles can be quoted as a multiple of PSA; for example, 200 PSA would be a vector with 0.4c to start and increase by 0.4c each month until reaching 12c in month 30. The name comes from the [Public Securities Association](#) deciding on this prepayment standard in 1985. Principal-bearing, floating rate bonds are quoted in discount margin (DM) terms to their index rate, reflecting the spread the bond offers above or below the reference rate. Interest-only derivatives are typically quoted in price terms and investors look to modeled yields and OASs as a starting point to these prepayment-sensitive products.

Returning to the structuring timeline, A CMO desk typically takes inquiry about customized structures or offers ideas to investors until a monthly deadline arrives for deciding on final deal structure. Fannie Mae, Freddie Mac, and Ginnie Mae set this deadline, known as the structured due date. For Ginnie Mae, the final structure date is around the 12th-15th, and Fannie Mae and Freddie usually set theirs a week after. At this point, the desk commits to underwriting a deal of the exact size and structure specified. The desk may have priced and sold individual classes of the deal before the structured due date, but at this point all details are final.

The collateral for the CMO can be held by the dealer or the client and is exchanged at the end of the month with the agencies for the CMO classes specified on the structure due date. Prior to the final CUSIP, dealers make the bonds available for analysis via preliminary “STRU” bonds. STRU allows clients to investigate the structure on Bloomberg before the end of the month. The final CUSIP is created when the deal is submitted to the relevant agency. The classes that have already been sold to investors then settle in the investors’ accounts, and unsold classes go into dealer inventory. From idea through collateral to structure, the deal is done.

IV. Stripdown pass-throughs

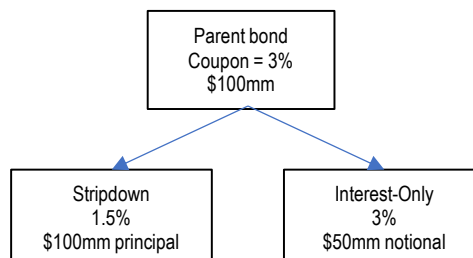
These structures are in the running for the simplest cashflows in CMO space, as they merely take all the principal and some of the interest from an existing bond and redirect the rest of the interest into an IO (**Figure 5**). They deserve mention due to their extreme popularity during the low-rate environment of 2020/2021. At that time, the entire mortgage universe was trading at a significant premium. FNCL 3s, the most issued coupon at the start of 2020 and functionally the lowest available coupon in 30yrs, priced above \$105 by late March. Additionally, banks were swamped with deposits from QE4 and looking for places to put it to work in MBS. The natural result was the creation of simple stripdown PTs that carried prices near par. At all times, these synthetic discounts can be structured further than just a PT (PACs are common) and are often created together with a floater (at a higher cap than the parent bond WAC) and an IIO.

Figure 4: Stripdown pass-through essentials

Key Features of Stripdown Pass-Throughs	Equivalent to a regular PT with a very high WAC
Common Uses	Reduce premium risk in very low rate environments

Source: J.P. Morgan

Figure 5: Structuring example for a stripdown PT

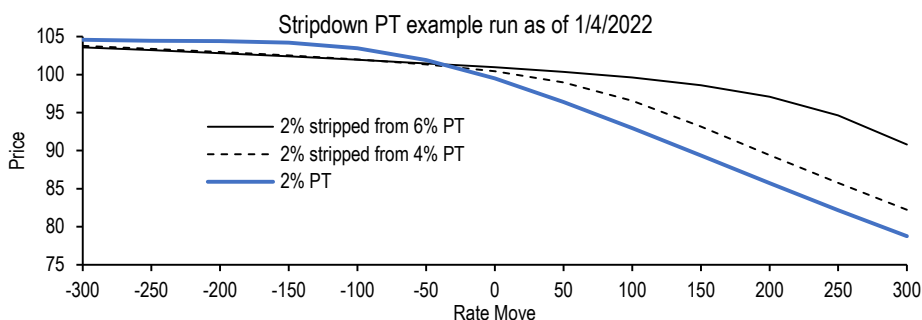


Source: J.P. Morgan

For valuation purposes, the stripdown PTs function like typical PTs with very high WACs. Compared to a regular PT with the same coupon, stripdowns typically have less duration around par as their collateral is in-the-money and prepaying faster. **Figure 6** demonstrates the price profile across rate moves using an FNCL 2, a 2% stripdown PT created from an FNCL 4, and a 2% coupon from an FNCL 6. Stripdowns can provide extension protection into a sell-off relative to the comparable coupon due to the higher prepayments in the underlying collateral.

Figure 6: The price profile of stripdown PTs relative to the comparable coupon PT

Pricing 2% stripdown PTs off 4% and 6% parent bonds (with normalized collateral) relative to a regular 2% PT, run at 0 Tsy OAS on 1/4/22



Source: J.P. Morgan

V. Sequentials

Sequential-pay classes, usually called sequentials, were among the first and most simple CMO structures created. Sequentials divide pass-through cash flows into multiple classes with different average lives and durations. A typical CMO sequential deal consists of three or more tranches carved out of the underlying collateral. Both scheduled amortization and prepayments from the collateral go to the holder of the first tranche until that tranche pays down completely. Principal then goes to the second tranche until that pays down. Then principal flows to each remaining class sequentially until each is retired and all pool principal has been repaid. Monthly interest from the pass-through goes to each class based on its principal outstanding.

Figure 7: Sequential essentials

Key Features of a Sequential-Pay Class	Targeted average life and duration
	Average life variability
	Principal lockout
	Principal window
Common Uses	Match the average life or duration of a liability
	Manage yield curve exposure
	Manage prepayment risk

Source: J.P. Morgan

This structure creates classes with different average lives and durations, but it also creates different principal lockout and principal windows for each class. Classes that have yet to receive principal are said to have principal lockout, and the period over which each class receives principal is called its principal window.

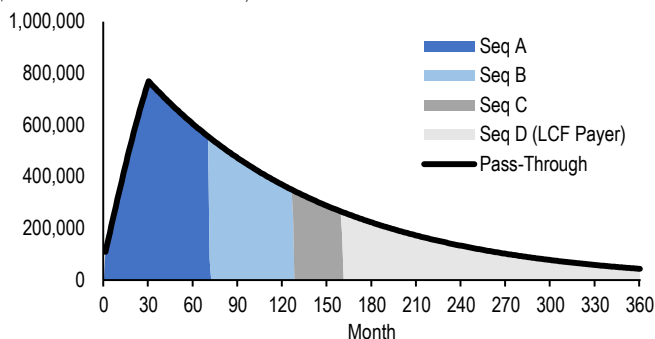
In all sequentials, the underlying collateral matters. The collateral determines the range of average lives and the lockout, principal windows, and prepayment sensitivities of the sequentials structured from it.

As an example, take the cash flows from a \$100-million pool of FNMA 6% 30-year pass-throughs. Unstructured, and assuming 150 PSA the pool offers investors the single choice of a bond with a 9.3-year average life. At different prepayment speeds, the average life could vary considerably. As a CMO split into four sequential classes, however, the investor can choose among different bonds with average lives either shorter or longer than the underlying collateral (**Figure 8** and **Figure 9**).

In this hypothetical sequential structure, class A receives the first \$40mm of principal, whether through amortization or prepayment. Class B receives the next \$25mm, and classes C and D receive the next \$10mm and \$25mm. In practice, the number of classes in a sequential structure and their initial principal balances depends on the pass-through used for collateral, the assumed prepayment speed, and investor demand.

Figure 8: Example of a sequential pay structure ...

\$100mm 6% PT at 6.85% WAC, run at 150 PSA



Source: J.P. Morgan

Figure 9: ... and the average life of the different tranches

\$100mm 6% PT at 6.85% WAC, run at 150 PSA

Tranche	Size (\$)	Average Life (years)
A	40,000,000	3.3
B	25,000,000	8.1
C	10,000,000	11.9
D	25,000,000	19.3
Pass-Through	100,000,000	9.3

Source: J.P. Morgan

Because of the rules for paying principal sequentially, all but the first class in this example initially have principal lockout and all have different principal windows. Although each class has an expected average life, the structure does not eliminate prepayment risk. If prepayments run faster than expected, all of the tranches' average

lives will shorten. And if prepayments come in slower, all tranches' average lives will extend. Sequential CMOs usually trade at a price and with a yield that reflects this prepayment risk.

Investors have used sequentials to match the average life or duration of MBS more closely to their liabilities. Portfolios with short or long liabilities, or money managers wanting to add exposure to the short or long part of the yield curve, can usually find a sequential with an appropriate average life and window. Principal lockout also adds some protection against spikes in prepayments, since earlier classes need to retire before an investor begins getting principal. Short classes buffer longer classes from prepayment risk.

Sequentials have attracted a wide audience. Banks in particular like the average life and yield of short classes. Money managers often invest in short or intermediate sequentials as substitutes for pass-throughs or ABS. Insurance companies often buy the last class of sequential deals for their long duration and yield, frequently comparing them with long credit and agency CMBS.

VI. Planned Amortization Classes (PACs)

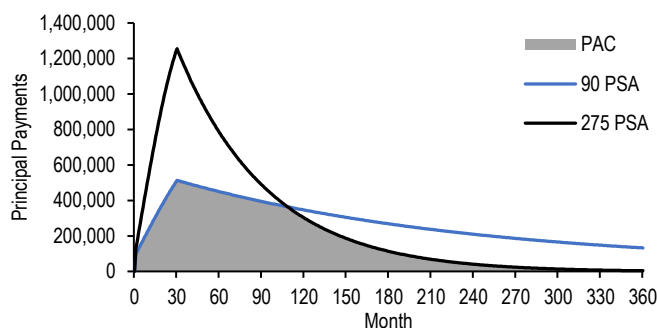
Although sequentials address investor interest in MBS with different average lives, some investors still find the variability of those average lives unmanageable. Another major class of structures—Planned Amortization Classes (PACs)—allows for different initial average lives but keeps those average lives stable over a range of prepayment rates and can be a way to shorten duration. Depending on the details of their structure, interest rates may rise or fall by varying amounts, but a PAC will still return principal on a predetermined schedule. A PAC bond may not only fit more closely to an investor's liabilities, but also may significantly reduce portfolio prepayment risk. This structure has found a wide audience among investors willing to trade off some yield in return for a more predictable return of principal and consequently a better price profile than a comparable sequential or pass-through.

Figure 10: PAC essentials

Key Features of a PAC Class	PAC bands
	Targeted average life and duration
	Principal lockout
	Principal window
	Collateral
Common Uses	Reduce portfolio prepayment risk
	Match the average life/duration of a liability
	Manage yield curve exposure

Source: J.P. Morgan

Figure 11: Creating a PAC principal repayment schedule taking the minimum of the two band schedules



Source: J.P. Morgan

PACs return an investor's principal according to a specific schedule as long as prepayments stay within a specific range, which is often referred to as a PAC collar or PAC band. The schedule is created by taking the minimum of two amortization schedules—the lower and the upper band of the PAC collar (**Figure 11**).

The rules that define a PAC bond distinguish between the PAC and the accompanying support. Any principal received goes first to paying down the PAC bond to its scheduled balance. If the principal received in any month falls short of the amount needed to keep the PAC on schedule, principal in later months still goes to the PAC until it returns to schedule. This is sometimes called a catch-up feature. Principal in excess of that needed to keep the PAC on schedule goes to paying down the support.

In structuring a PAC, the support absorbs an important part of the prepayment risk. Consider our example PAC structured using 90 PSA as the lower bound and 275 PSA as the upper bound. At speeds within the PAC collar, the average life of the untranched PAC bond—often called the PAC strip—remains constant at 7.7 years, while the average life of the support fluctuates widely—for example, from 18.7 years at 100 PSA to 4.7 years at 250 PSA (**Figure 12**).

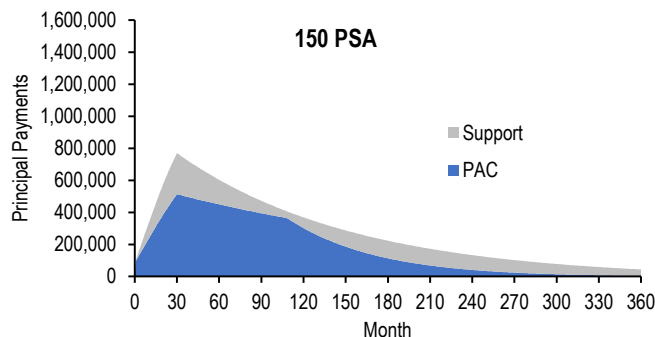
Figure 12: Average life of PAC strip at different prepayment speeds (90/275 PSA collar)

Speed	PAC (90/275 band)	Support
50 PSA	9.8	24.3
100 PSA	7.7	18.7
150 PSA	7.7	12.4
200 PSA	7.7	7.9
250 PSA	7.7	4.7
300 PSA	7.3	3.0
400 PSA	5.9	2.2

Source: J.P. Morgan

Figure 13: PAC and support payments at 150 PSA

\$100mm 6% PT (6.85 WAC), 90/275 PSA collar



Source: J.P. Morgan

The support absorbs prepayment variability to give stability to the PAC class. As prepayments rise, the support gets payment in excess of the amounts required to keep the PAC on schedule, thus shortening the support's average life. As prepayments fall, the support extends again. **Figure 13** shows the typical cashflow on a PAC strip/support at a speed within the PAC collar.

Over time, the collar on a PAC changes as actual prepayments unfold. The types of changes to the PAC collar fall into three categories:

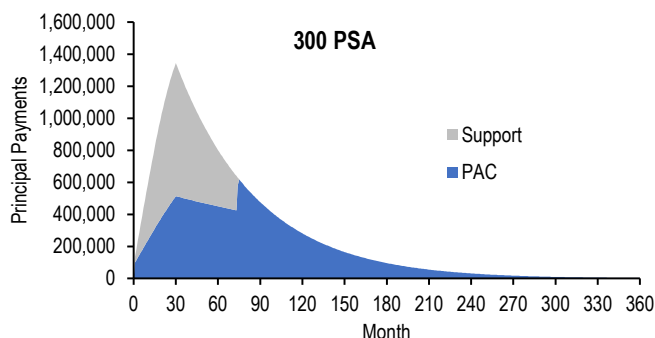
- *Speeds Above the PAC Collar (Figure 14).* At prepayment speeds above the upper band of the PAC collar, the support begins to pay off quickly. With less support principal outstanding, the structure begins to lose its ability to keep the PAC on schedule at the original upper PAC band. The upper band begins to drift down. The drop in total outstanding principal also lowers the flow of monthly principal that comes from simple amortization and consequently weakens the structure's ability to keep the PAC on schedule at the original lower PAC band. The lower band begins to drift up. Overall, the PAC collar narrows from both the upper and lower direction. In the extreme, if all support principal prepays, the PAC structure breaks and the investor is left with a cash flow almost identical to a pass-through.
- *Speeds Within the PAC Collar.* At prepayment speeds close to the upper band of the PAC collar, the support class pays off at a relatively fast rate. The upper band stays in place since enough support principal remains to pay the PAC and divert any extra monthly principal to the support. But the lower PAC band drifts up. Less and less

principal remains to provide the amortization required for the PAC to stay on schedule at slow speeds. At speeds close to the lower band of the PAC collar, however, the deal has relatively more principal outstanding. This helps keep the PAC on schedule at speeds above the original upper band and below the original lower band. In this case, the PAC collar widens from both directions.

- **Speeds Below the PAC Collar (Figure 15).** At prepayment speeds below the lower band, the amount of principal outstanding is larger than expected, but any PAC currently due to receive principal falls behind schedule. The upper band drifts higher because more support bonds remain to absorb spikes in prepayments. The behavior of the lower band depends on whether the PAC is currently due to receive principal. PACs currently due principal fall behind schedule, so the lower band drifts higher since the deal needs to prepay faster for the PAC to catch up to its schedule. For PACs currently locked out from receiving principal, however, the lower band drifts down. This widening of the collar for locked out PACs reflects the greater amount of support bonds available to protect against both fast and slow prepayments.

Figure 14: Speeds above the upper collar erode the support ...

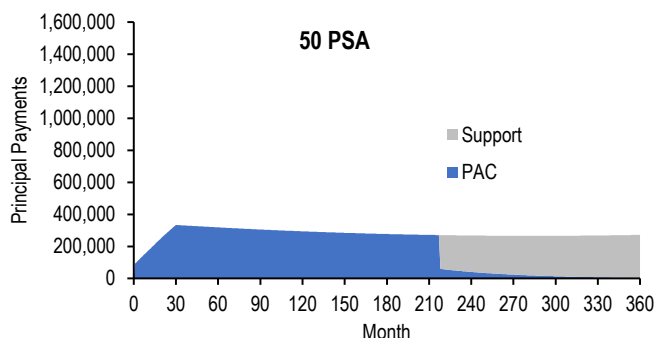
\$100mm 6% PT (6.85 WAC) , 90/275 PSA collar at 300 PSA



Source: J.P. Morgan

Figure 15: ... and speeds below the lower collar extend the PACs

\$100mm 6% PT (6.85 WAC) , 90/275 PSA collar at 50 PSA



Source: J.P. Morgan

Just like a sequential structure, a PAC can be divided into tranches with different average lives. Both scheduled amortization and prepayments from the underlying collateral go to the first PAC until it is retired, to the second PAC until it is retired and so on until all the PACs pay off. Monthly interest payments from the pool are apportioned to the classes based on outstanding principal.

PACs add prepayment protection to a portfolio since principal gets returned according to schedule as long as the support classes remain outstanding. PACs have many of the same uses as sequentials but often work better for portfolios that need more predictable principal return and/or shorter duration. No PAC can make principal return perfectly predictable, of course. But PACs usually have more stability than pass-throughs or sequentials.

PACs attract a wide audience but get particular attention from money managers. PACs' relative stability can make the structure a good substitute for U.S. Treasuries, agencies, and corporate bonds. And PACs allow money managers to take positions on the yield curve more efficiently than pass-throughs or sequentials. Money managers typically buy 7-year and shorter structures. Banks use PACs periodically to add prepayment protection, usually focusing on 5-year and shorter classes. The 10- and 20-year PACs often trade to insurance companies.

VII. Floaters

Not all investors want to take the interest rate risk associated with a security that has a fixed coupon, and for those investors the CMO market has developed floating-rate classes. Floating-rate CMOs have coupons that reset periodically to reflect changing interest rates—at least as long as rates remain below a predetermined maximum or cap. The floating coupon tends to keep the price of the security around par while factors such as margin and distance from the cap can move the price above or below \$100. This structure can have tremendous value for investors that want to limit volatility in the price of their MBS (by limiting duration), or for investors that finance their MBS using floating-rate funds.

Figure 16: Floater essentials

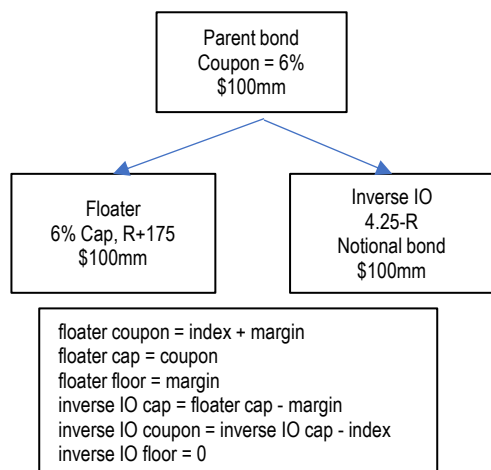
Key Features of Floating-Rate Classes	A floating-rate coupon based on floating-rate index + margin
	A cap on the coupon
	The underlying fixed-rate class
Common Uses	Limit interest rate risk
	Earn a spread to floating-rate funding
Common Features	Structure = the underlying fixed-rate structure such as a pass-through, sequential, PAC, etc.
	Index = the reference rate, such as 1-month LIBOR or 30 day average SOFR
	Margin = spread of the floater over the index
	Cap = the maximum value of the floater
	Strike = the level of the index where the coupon hits its cap
	Delay = The number of days between the end of the accrual period for the floater coupon and the payment of the coupon, such as 0- or 24-day delay
	Discount Margin = The difference between the yield of the floater and its index, used to estimate the effective margin for floaters priced above or below par

Source: J.P. Morgan

Structuring a floater always starts with an MBS that has a fixed coupon. That underlying MBS can take almost any form—pass-through, sequential, PAC, TAC, or supports.

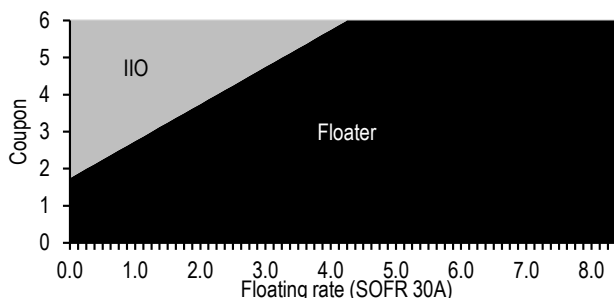
Figure 17 shows the most basic method of creating a floater, and **Figure 18** shows how the original pass-through's coupon would be divided between the floater and the inverse IO for this same method.

Figure 17: Structuring a floater



Source: J.P. Morgan

Figure 18: Original 6% pass-through's coupon split across a floater (6-cap, 175 DM) and an Inverse IO



Source: J.P. Morgan

When constructing a floater, the most important consideration for creating the cap is that the total principal/interest going to the child classes must always equal the principal/interest for the parent bond. These considerations create the floating-rate CMO's cap. A dealer could take a \$100 million pass-through or CMO class with a 6.0% coupon, for example, and create a \$100 million floater. In this instance, the maximum possible coupon or cap for the floater would be 6.0%. There are ways to divert the interest and principal of the initial/parent bond to raise that cap, which we will expand on in the following section.

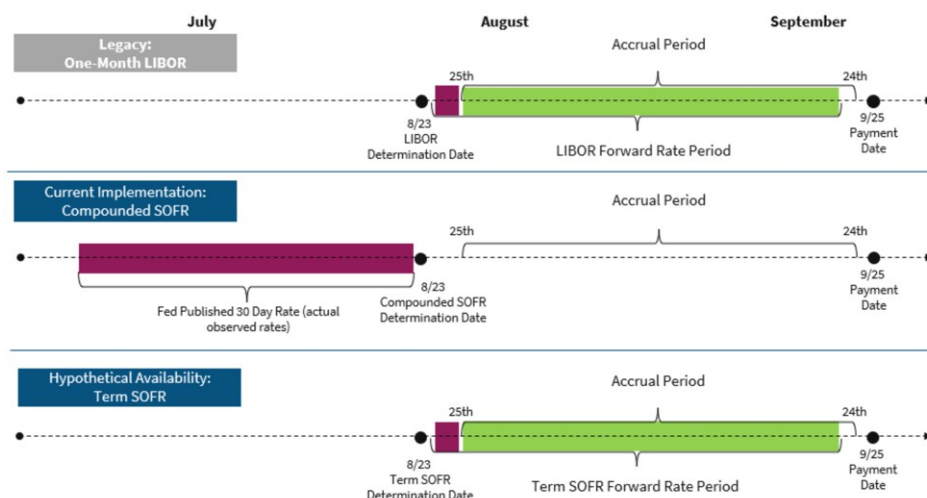
The other key features of a floater are its index and margin. The index is the reference interest rate that determines how and how often the floater coupon will change. After many years of most structures referencing 1m LIBOR, the new standard is 30 day compounded average SOFR, available on the [NY Fed's website](#). The margin is the spread of the coupon over the index. A floater based on 30d SOFR with a 175 basis point margin would have a coupon that resets monthly to S+175bp. The coupon would reset according to this formula as long as it was less than or equal to the floater's cap. The value of the index where the floater hits its cap is called the floater's strike. The floating coupon would bottom out at its margin (175bp in this example) when 30d SOFR reaches 0% or lower.

Some CMO floater investors that finance their securities would have problems if the floater coupon arrived only after the 19- or 24-day delays common with pass-throughs or most CMOs. The delay creates a mismatch between receiving the floater coupon and paying for the floating-rate financing. As a result, the vast majority of floaters are structured by dealers as 0-day delay. These floaters begin accruing their coupon on the 20th or 25th day of the month and stop accruing on the 14th or 24th day of the next month, paying their coupon the day afterwards. This way, investors can settle both their purchase and financing on the same day and cover financing costs without any mismatch in timing. **Figure 19** gives a sense of the timeline between rate determination and payment using both the old LIBOR framework and the new compounded SOFR

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system.

Figure 19: Timeline of floater payments (with 0 delay, green/red shows where the rate does and doesn't align with the accrual period, reproduced from the [LIBOR Transition Playbook](#))



Source: J.P. Morgan, Fannie Mae, Freddie Mac

VIII. How do dealers lift floater caps above the collateral coupon?

In certain environments, investors may prefer higher-cap floaters to reduce cap risk, particularly in periods of rising current-coupon yields and elevated rate volatility. Dealers can structure higher caps than the coupon of the collateral pools in several ways, which we examine below. The choice of method depends on demand for the following higher-cap floater offshoots:

- Stripdown PT (and further cuts),
- Inverse floater,
- Mountain, or
- TTIBs

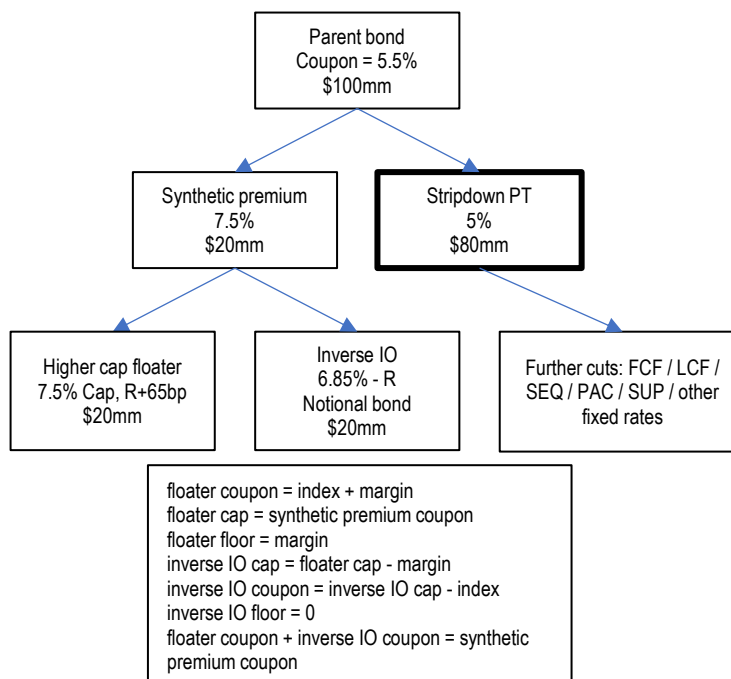
Alternatively, the GSEs can help gross up (or even uncap) floaters through:

- IRCs

Stripdown PT (and further cuts)

As shown in **Figure 20**, a floater's cap can be raised above the collateral coupon by using a portion of the parent bond's principal to create a synthetic premium. The offsetting synthetic discount, like any stripped-down pass-through, can then be cut further.

Figure 20: Structuring a high cap floater through a Stripdown PT (including further cuts)



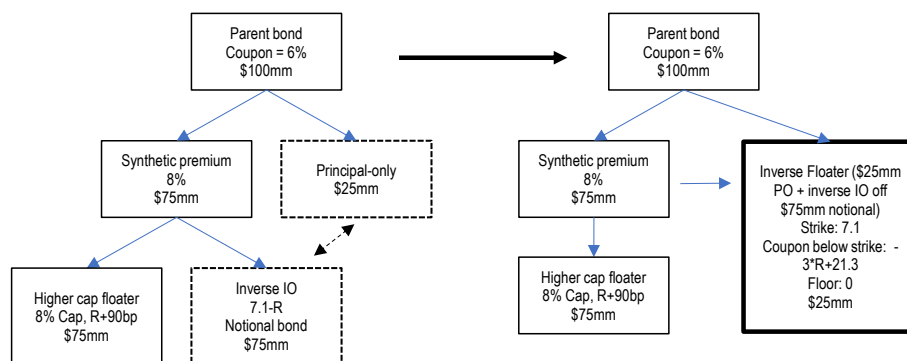
Source: J.P. Morgan

As discussed in the main stripdown pass-through section, these structures were popular in the low-rate environment of 2020–2021, when banks had significant duration needs and sought to minimize premium risk. As of this writing though, stripdowns trade at a discount to par given the coupons being created, so dealers have since explored other ways to raise floater caps to meet different investor needs.

Inverse floater

Another way dealers raise floater caps is by creating inverse floaters. As shown in a theoretical example in **Figure 21**, inverse floater creation begins in the same way as PO or stripdown creation; instead of further cutting the original stripdown though, the inverse floater combines the fixed rate with the inverse IO created off the synthetic premium.

Figure 21: Structuring a high cap floater through an inverse floater (through a PO / stripdown PT)

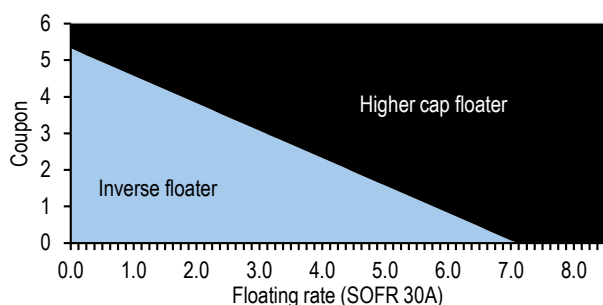


Source: J.P. Morgan

If an inverse floater attaches an inverse IO off a larger notional to a PO of smaller size, it will be levered; if its inverse IO notional is smaller than the PO's size, its sensitivity to SOFR will be less than its companion floater; and if its inverse IO notional is equal to the PO's size, it will have the same sensitivity to SOFR as its companion floater (**Figure 22 & Figure 23**). In other words, this leverage is dependent on the relative size of the synthetic premium used to create the higher-cap floater and the inverse IO, and the PO/stripdown. This is, in turn, dictated by how much higher a dealer raises the cap above the original pass-through's coupon as well as the coupon of the stripdown. Our theoretical example splits the parent bond into a synthetic premium and a PO; creating a stripdown with a coupon instead of a pure PO adds a floor above 0 to the resulting inverse floater.

Figure 22: If an inverse floater attaches an inverse IO off a larger notional to a PO of a smaller size ...

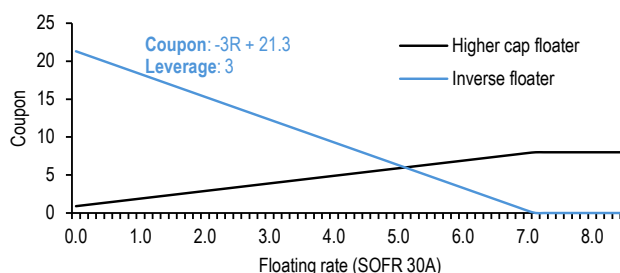
Original 6% pass-through's coupon split across an inverse floater and an 8-cap (90 margin) floater



Source: J.P. Morgan

Figure 23: ... then its coupon is levered below its strike

Coupon (%) vs. floating rate (SOFR30A, %) for a theoretical 8-cap (90 margin) floater and inverse floater created off a 6% pass-through



Source: J.P. Morgan

Historically, inverse floaters were created so that investors could effectively take a leveraged position in an underlying fixed-rate pass-through. This was particularly popular if funding a pass-through position through this method was more attractive than other forms of financing. Over the past several years though (2023-2026), this need has eased as balance sheet capacity has grown. Instead, inverse floaters are now a way for investors to express views on speeds, curve, and/or duration.

Figure 24: Inverse floater summary

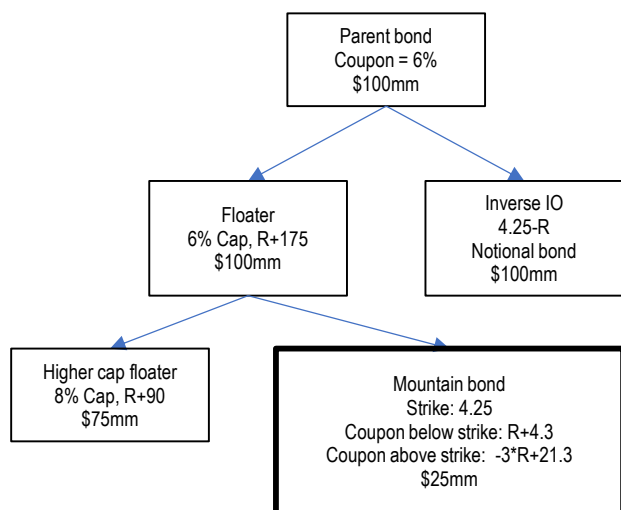
Key Features of Inverse floaters	Strike (equal to companion floater cap - companion floater margin) Floating rate coupon below strike: below-strike slope * floating-rate index + below-strike margin Leverage (-1 * below-strike slope) is equal to [Higher-cap floater size] / [Inverse floater size] where Higher-cap floater size is equal to the (original coupon / higher-cap) * original pass-through size Floor: Based on how much fixed-rate is allocated to the stripdown opposite the higher-cap floater and is equal to 0 if the stripdown opposite the higher-cap floater is a PO
Common Uses	Expressing speeds, curve, and/or duration view Levered view on pass-through (though less common in recent years as balance sheet capacity has grown)

Source: J.P. Morgan

Mountain

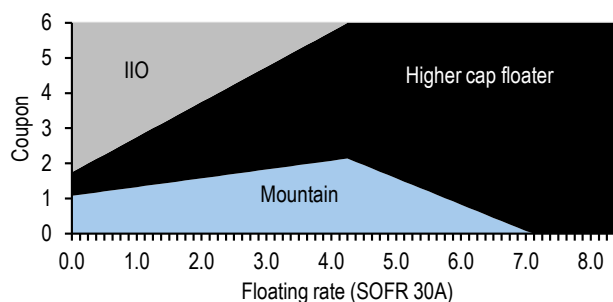
Another way to raise a floater's cap is to create a higher-cap floater and a mountain bond from an original, lower-cap floater, with the principal allocated to each dictated by how much the dealer wants to increase the cap. **Figure 25** shows a theoretical mountain and an 8-cap floater created from a 6-cap floater, and **Figure 26** shows how the original parent bond's coupon in our theoretical example is allocated across its mountain, higher-cap floater, and inverse IO pieces across different floating rates.

Figure 25: Structuring a high cap floater through a mountain



Source: J.P. Morgan

Figure 26: Original 6% pass-through's coupon split across a mountain created off a 6-cap (175 margin) floater, an 8-cap (90 margin) floater, and an Inverse IO



Source: J.P. Morgan

A mountain bond's coupon varies with its benchmark floating-rate index. Like its parent floater, its coupon equals the floating rate plus a margin up to its strike (which always matches the strike of the inverse IO in its group), at which point the coupon peaks. Unlike its parent floater, however, a mountain's coupon declines linearly above the strike at a levered pace until it reaches its floor. In our theoretical example, the mountain bond's leverage is 3.

Figure 27: Mountain summary

Key Features of Mountains	Strike (equal to I/O strike in the same group)
	Floating rate coupon below strike: floating-rate index + below-strike margin
	Inverse floating rate coupon above strike: above-strike slope * floating-rate index + above-strike margin
	Leverage (-1 * above-strike slope) is equal to [Higher-cap floater size] / [Mountain bond size] where higher-cap floater size is equal to the (Original cap / higher-cap) * original floater size
Common Uses	Absorbs residual cap risk (short volatility)
	Expressing a specific short rate view (with leverage above strike)

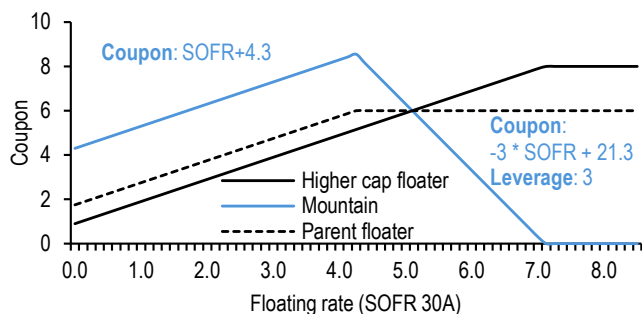
Source: J.P. Morgan

Figure 28 illustrates how the mountain in our example varies with rate. In effect, the mountain absorbs the marginal cap risk between the parent floater and the higher-cap floater. Investors who purchase mountain bonds express a specific rate view and are long this cap risk; in doing so, they are short gamma and vega.

For a given parent floater, a mountain's below-strike margin, above-strike margin, and leverage all vary with its companion high-cap floater's cap and corresponding margin. Mountain leverage depends on its companion floater's cap relative to the original (lower) cap: as the amount by which a dealer boosts the cap decreases, the corresponding mountain's leverage grows hyperbolically (**Figure 29**). In theory, a mountain would no longer have leverage if its companion floater's cap was at least double the original, but it is difficult to find mountains like this.

Figure 28: Mountains absorb the marginal cap risk between the parent and higher-cap floaters ...

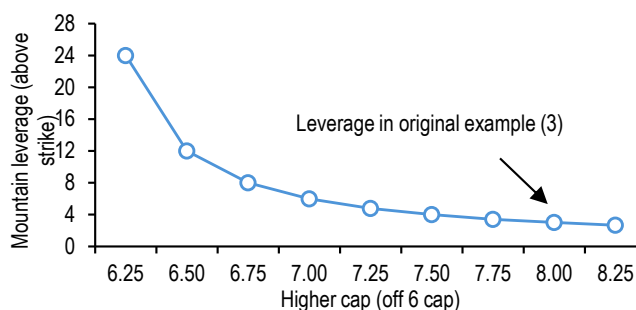
Coupon (%) vs. floating rate (SOFR30A, %) for a theoretical 8-cap (90 margin) floater and mountain created off a 6-cap (175 margin) floater



Source: J.P. Morgan

Figure 29: ... and their leverage is a function of how much a dealer increases cap by

Mountain leverage (the negative slope above strike) for theoretical mountains off a 6-cap (175 margin) floater across a range of companion higher-cap floater caps, %



Source: J.P. Morgan

Figure 30: Mountain attributes are a function of the amount by which a dealer increases cap, as well as what that cap's DM is at the time of creation

Mountain margins above and below strikes for theoretical mountains off a 6-cap (175 margin) floater across a range of companion higher-cap floater caps and a range of DMs on these same higher-cap floaters, %

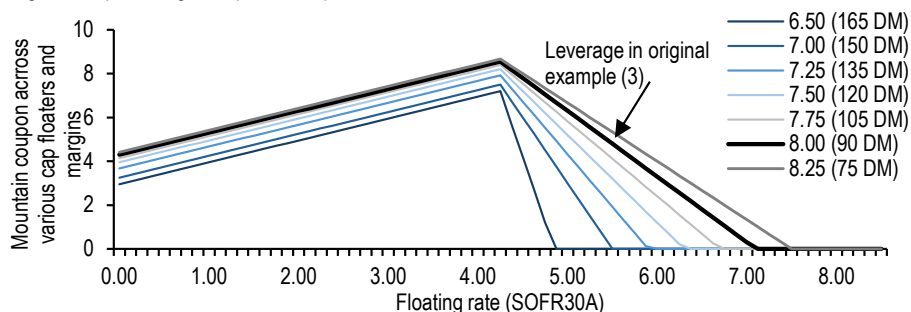
	Margin (below strike, %)							Margin (above strike, %)						
	165	150	135	120	105	90	75	165	150	135	120	105	90	75
6.25	4.15	7.75	11.35	14.95	18.55	22.15	25.75	110.4	114.0	117.6	121.2	124.8	128.4	132.0
6.50	2.95	4.75	6.55	8.35	10.15	11.95	13.75	58.2	60.0	61.8	63.6	65.4	67.2	69.0
6.75	2.55	3.75	4.95	6.15	7.35	8.55	9.75	40.8	42.0	43.2	44.4	45.6	46.8	48.0
7.00	2.35	3.25	4.15	5.05	5.95	6.85	7.75	32.1	33.0	33.9	34.8	35.7	36.6	37.5
7.25	2.23	2.95	3.67	4.39	5.11	5.83	6.55	26.9	27.6	28.3	29.0	29.8	30.5	31.2
7.50	2.15	2.75	3.35	3.95	4.55	5.15	5.75	23.4	24.0	24.6	25.2	25.8	26.4	27.0
7.75	2.09	2.61	3.12	3.64	4.15	4.66	5.18	20.9	21.4	21.9	22.5	23.0	23.5	24.0
8.00	2.05	2.50	2.95	3.40	3.85	4.30	4.75	19.1	19.5	20.0	20.4	20.9	21.3	21.8
8.25	2.02	2.42	2.82	3.22	3.62	4.02	4.42	17.6	18.0	18.4	18.8	19.2	19.6	20.0

Source: J.P. Morgan

Relatedly, the smaller the difference between the higher cap and the original, the higher the mountain's above-strike margin — and, for tighter higher-cap margins, the higher the below-strike margin as well (**Figure 30**). In short, tighter higher-cap floater margins raise a companion mountain's below- and above-strike margins, though to varying degrees depending on the cap. This is shown in **Figure 31** on theoretical mountains created from a 6-cap 175 margin parent floater across a range of companion higher-cap floaters bearing various margins. These margins are illustrative only and do not necessarily reflect real-time levels at publication.

Figure 31: Theoretical mountains created off a 6-cap parent floater with respect to a range of companion higher-cap floaters bearing various margins

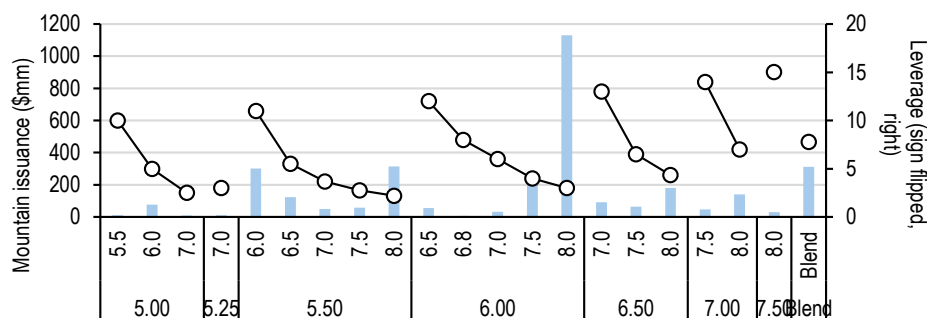
Mountain coupon (%) vs. floating rate (SOFR30A, %) for theoretical mountains off a 6-cap (175 margin) floater across a range of companion higher-cap floater caps and their DMs



Source: J.P. Morgan

In practice, the maximum mountain leverage we have observed hovers below 16, and the bulk of issuance has come from 6-cap collateral paired with 8-caps, yielding a leverage of 3 (**Figure 32**). Mountain bonds with more leverage than this are less common because dealers have some flexibility in choosing the collateral coupon or original cap from which to create mountains.

Figure 32: Mountain issuance from 2023 onwards (left, \$mm) and leverage (sign flipped) split by parent floater cap / companion floater cap, for large parent floater caps



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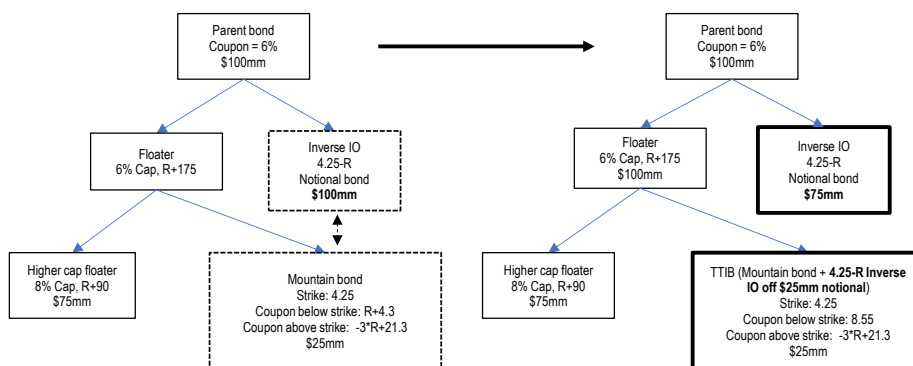
Figure 33: TTIB summary

Key Features of TTIBs	Strike (equal to IIO strike in the same group) Coupon below strike: Strike + below-strike margin (constant) Inverse floating rate coupon above strike: above-strike slope * floating-rate index + above-strike margin Leverage (-1 * above-strike slope) is equal to [Higher-cap floater size] / [TTIB bond size] where higher-cap floater size is equal to the (Original cap / higher-cap) * original floater size
Common Uses	Absorbs residual cap risk (short volatility) Expressing a (levered) long duration view (in the range between the IIO strike and the higher-cap floater's strike) while expressing the view that the structure's inverse IO is attractive

Source: J.P. Morgan

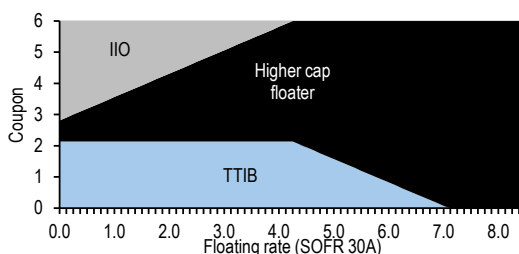
As shown in **Figure 34**, a TTIB is a mountain bond plus an inverse IO drawn from the parent floater's companion, scaled to the mountain bond's notional. This makes a TTIB's coupon equal to the original mountain's peak for floating rates below its strike. **Figure 35** shows how the original parent bond's coupon in our theoretical example is allocated across its TTIB, higher-cap floater, and IIO pieces across rates.

Figure 34: Structuring a high cap floater through a TTIB (through a mountain)



Source: J.P. Morgan

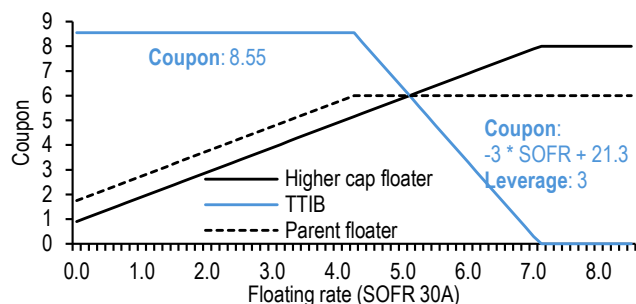
Figure 35: Original 6% pass-through's coupon split across a TTIB created off a 6-cap (175 margin) floater and 25% of its companion inverse IO, an 8-cap (90 margin) floater, and an Inverse IO



Source: J.P. Morgan

Figure 36: TTIBs have a steady coupon below strike and thus, more stable carry ...

Coupon (%) vs. floating rate (SOFR30A, %) for a theoretical 8-cap (90 margin) floater and TTIB created off a 6-cap (175 margin) floater and 25% of its companion inverse IO



Source: J.P. Morgan

Figure 37: ... so they generally pick more OAS than their equivalent mountain bond

Tsy OAS for a theoretical mountain created off a 6-cap (175 margin) floater opposite an 8-cap floater, for 25% of the parent floater's companion inverse IO, and for the combination of the two, or the implied TTIB, bp

	Tsy OAS (bp)
Mountain	38
Inverse IO (same notional)	958
Combined (implied TTIB)	48

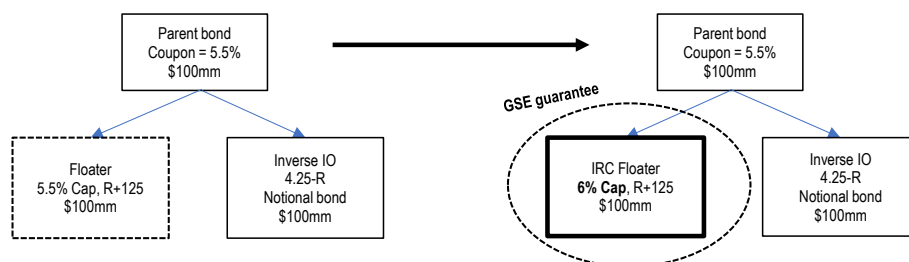
Source: J.P. Morgan

As shown in **Figure 36**, a TTIB's coupon is steady below its strike: although a TTIB still absorbs the cap risk needed to raise its companion floater's cap, its investor takes on a less specific rate view than a mountain investor. Therefore, TTIBs offer a more stable carry profile than mountains and pick more OAS than mountains because of its inverse IO component; as shown in **Figure 37**, if we simply combine the mountain and an equivalent notional inverse IO in the same group, and run them as a TTIB, the combined structure picks more OAS than the mountain. TTIBs tend to be more popular than mountains when the inverse IO in that same group is considered inexpensive.

Interest rate cap (IRC)

Depending on their willingness to assume some marginal cap risk, the GSEs can help raise floater caps by guaranteeing the higher-cap floater, as shown in **Figure 38**.

Figure 38: Structuring a high cap floater through an IRC guarantee



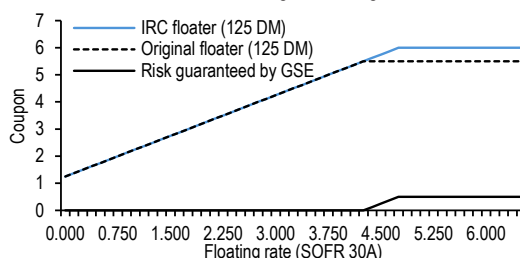
Source: J.P. Morgan

Through the GSE guarantee, the IRC floater can maintain the same margin as its original floater despite its higher cap, provided the GSE assumes all of the marginal cap risk between the old and new strikes (**Figure 39**). The GSE is compensated for taking on this cap risk by earning the premium on the IRC (priced at par) versus the original floater (priced at a discount to par). This premium is valued by running the IRC and original floater at even OAS through a prepay model, to account for the option cost from both rates and speeds affecting how the interest rate cap amortizes over its term.

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Figure 39: A GSE can offer a higher-cap floater with the same margin as the original un-guaranteed floater by shouldering the marginal cap risk, for a price

Coupon (%) vs. floating rate (SOFR30A, %) for a IRC floater (125 margin) as well as its breakout into its original floater and the risk the GSE takes on through the IRC guarantee



Source: J.P. Morgan

IX. Supports

While many investors find the stability of sequentials, PACs and TACs useful, others have looked for ways to earn additional yield in agency MBS by accepting more exposure to prepayments. Especially for portfolios needing current income, taking on or leveraging prepayment risk in agency MBS is one of the few ways in U.S. fixed income of adding income without adding credit risk. These investors often turn to support cash flows.

Figure 40: Support essentials

Key Features of Supports	High yield
	Average life variability
	Principal window
Common Uses	Maintain portfolio yield
	Express a view on prepayments

Source: J.P. Morgan

In structuring stable cash flows for both PACs and TACs, the monthly instability from prepayments shifts to the supports. At fast prepayment speeds, the support classes absorb any principal in excess of that needed to keep the PACs or TACs on schedule. Average life can shorten dramatically. And at slow prepayment speeds, the supports go to the end of the line in the list of classes receiving principal payments. Average life can extend substantially. Because support classes leverage the prepayment risk in the underlying collateral, the collateral itself becomes one of the most important features of the structure.

Just like a PAC strip or TAC strip, a support can be structured into different tranches. The possibilities are diverse:

- *Support Sequentials.* This approach simply divides up the support into separate tranches that pay down sequentially. The first support class receives principal until it expires, then the second receives principal until it expires, and so on. These classes will differ in lockout, principal window, and average life. And they differ too in average life variability and yield.
- *PAC IIs.* This approach creates a set of tranches out of the support that get principal according to a schedule defined by a PSA collar—similar to a normal PAC I—but gets that principal only after any PAC I in the deal. A PAC II typically has a narrower collar than a PAC I. A PAC II might have a collar of 125-225 PSA in a deal

where the PAC I has a collar of 100-250 PSA. Principal goes first to the PAC I, then to the PAC II and then to any other support classes. If the supports pay off, then the PAC IIs begin to behave like support sequentials to provide prepayment protection for the PAC I.

- *Support TACs.* This structure creates a TAC out of the support. A single prepayment speed defines the support TAC's schedule, and principal goes first to pay the TAC down to its scheduled balance. Any excess goes to any other support classes. A support TAC generally breaks faster than a normal TAC because the support TAC gets its protection from a smaller set of support classes. And a support TAC generally extends more than a normal TAC because of the greater extension risk in the overall support. As usual, a support TAC can be structured to have different lockout, principal windows, and average lives.
- *Others.* The other structures carved out of support cash flows range from FFIEC bonds to support Z bonds, support floaters, to a host of other MBS derivatives

The presence of PAC IIs or support TACs in a structure means that any remaining support classes are even more sensitive to prepayments than the original support.

Support classes in their various forms usually go to investors trying to maximize portfolio yield or who have a view on collateral. After all, supports tend to have greater average life variability than sequentials, PACs, TACs or the underlying pass-throughs—and consequently have higher yield. For portfolios that expect interest rates and prepayments to be stable, supports can be excellent vehicles for capitalizing on that stability.

X. TACs

For some investors, PACs add too much stability to mortgage cash flows at the expense of yield. And sequentials leave too much risk on the table. For these investors, a Targeted Amortization Class or TAC provides protection against faster but not slower prepayments. This gives the TAC more negative convexity than a PAC because the TAC extends more as interest rates rise. Because it has less protection than a comparable PAC, a TAC typically trades at a higher yield. TACs have attracted investors that want to give up some extension protection in return for higher yield.

Figure 41: TAC essentials

Key Features of a TAC Class	TAC speed
	Targeted average life
	Principal lockout
Common Uses	Principal window
	Reduce MBS call risk
	Match the average life or duration of a liability
	Maintain/increase portfolio yield

Source: J.P. Morgan

Figure 42: Average lives of TAC tranches and support at different levels of prepayments (150 PSA TAC speed)

\$100mm 6% PT (6.85 WAC), shading shows nearly constant WAL

Speed	TAC	Support
100 PSA	9.2	25.3
125 PSA	8.0	23.8
150 PSA	7.1	22.1
175 PSA	7.2	16.1
200 PSA	7.3	10.7
225 PSA	7.5	5.3
250 PSA	7.3	2.7
300 PSA	6.5	2.0
350 PSA	5.7	1.7
400 PSA	5.2	1.5

Source: J.P. Morgan

Like PACs, TACs repay principal according to a schedule. Unlike PACs, however, the TAC redemption schedule is created from a single prepayment speed rather than the minimum of two speeds. If prepayment speeds exceed the one used to define the TAC schedule, the excess principal goes to the support class. If prepayment speeds fall below the speed used, both the TAC and the support class extend (**Figure 42**).

Just like sequentials and PACs, TACs can be further divided into tranches with different average lives. Each TAC pays down fully before the next tranche gets any principal. Monthly interest payments from the underlying collateral are apportioned to each class based on outstanding balance.

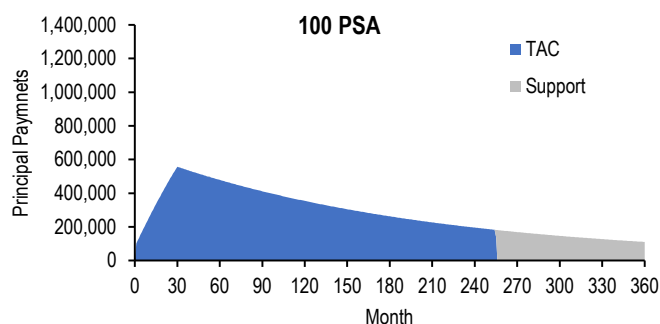
As with PACs, the cash flows in a TAC depend on the path of interest rates and the prepayments over the life of the security. A few types of scenarios are worth mentioning:

- *Speeds Below the TAC Speed (Figure 43).* At speeds below the TAC speed, both the TAC and support classes extend. If speeds remain slow for long enough, the growing volume of monthly amortization may eventually allow the TAC to catch up and then remain on schedule. Support tranches are structured to receive payments only after the TAC schedule is satisfied. At sustained slow speeds, more supports remain outstanding than projected at the TAC speed. The TAC may develop a narrow collar—like a PAC—where speeds above the TAC speed still allow the structure to remain on schedule.
- *Speeds Well Above the TAC Speed (Figure 44).* With sustained fast prepayments, the class supporting the TAC pay off and the TAC receives principal just like a pass-through. In this situation, the TAC breaks and begins to receive principal much earlier than scheduled. The TAC's weighted average life shortens.
- *Speeds Moderately Above the TAC Speed.* With prepayments only moderately faster than the pricing speed, the support helps maintain the schedules of TAC by absorbing excess cash flows, leaving it to extend very modestly if at all. Once the support pays off, prepayments may continue coming in quickly and the principal payments to the TAC end up being less than those specified in the TAC schedule. The TAC may end up taking in principal after its originally scheduled maturity, and

its average life extends. In a tranching TAC, the range of prepayment speeds where the TAC remains on schedule is wider for short than for long TACs.

Figure 43: TAC and support principal payments below the TAC speed at 100 PSA

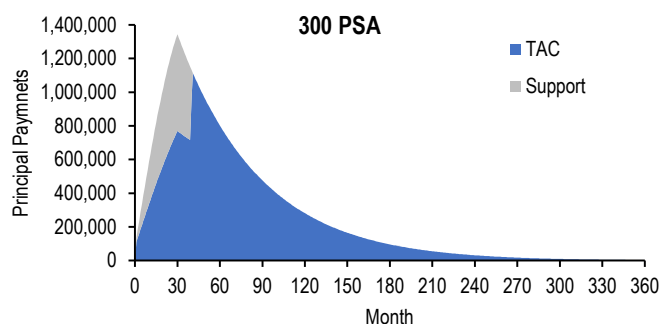
100mm 6% PT (6.85 WAC), 150 PSA TAC speed



Source: J.P. Morgan

Figure 44: TAC and support principal payments above the TAC speed at 300 PSA

100mm 6% PT (6.85 WAC), 150 PSA TAC speed



Source: J.P. Morgan

TACs attract investors that want better yield than afforded by a PAC but more call protection than afforded by a sequential. Classes structured off pass-throughs trading well below par sometimes offer relatively stable cash flows. In the underlying collateral, modest movements in interest rates have only small effects on prepayments. And the TAC structure helps protect against moves that spur fast prepayments. As with sequentials and PACs, TACs with different average lives also let portfolios line up assets more closely with liabilities. And principal lockout and window help manage prepayments and positioning along the yield curve.

XI. LCFs: Payers and Z Bonds

In a structure with sequential-pay tranches, the simplest being a sequential, the last tranche is often referred to as a last cash flow (LCF). A LCF that pays interest during the period of principal lockout is known as a LCF payer. Another popular structure is a Z bond, where interest from the LCF is directed to other tranches in the structure to pay them off faster. LCF payers look just like other tranches of the same structure (just longer) and in this section we will explore the unique features of Z bonds.

Z bonds attempt to meet the needs of investors looking for long duration and yield and deliver some of the highest levels of both that the mortgage market offers. Investors can find some duration and yield in long sequentials or PACs or other structures. But Z bonds create duration and yield by initially deferring payment of their coupon. This is the origin of their name, since they often get compared to zero-coupon Treasuries and other bonds.

Figure 45: Z bond essentials

Key Features of Z-Bonds	A zero-coupon class until the principal window opens
	Rising principal balance
	The underlying structure — sequential, PAC, companion or other
Common Uses	Add to portfolio duration

Source: J.P. Morgan

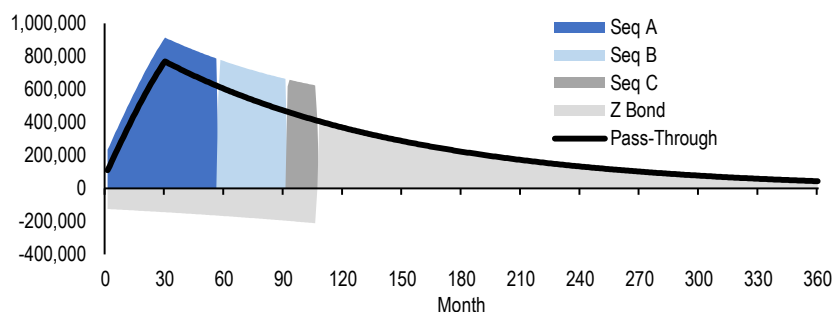
The cash flow of a Z bond falls into two phases:

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- **Accrual Phase.** During the accrual phase, the Z bond gets no interest payments. Interest due to the Z instead gets redirected toward paying down the outstanding balance of other tranches. The Z bond is credited for the forgone interest by increasing its principal balance each month by the amount of foregone interest. The Z's principal balance grows over time at the stated coupon rate, compounding at the payment frequency and avoiding any reinvestment risk. This accrual mechanism continues until all the bonds receiving the foregone interest have paid off. At that point, the accrual phase ends.
- **Payment Phase.** Once the bonds receiving the Z's foregone interest have paid off, the Z bond begins to receive both monthly principal and interest until it too is paid down.

Figure 46: Principal Cash Flows at 150 PSA

Principal repayment at 150 PSA for a 100mm 6% PT (6.85 WAC)



Source: J.P. Morgan

The effect of adding a Z bond to a sequential deal helps illustrate the Z structure. Principal payments to the sequential tranches—principal in excess of that normally received from the underlying collateral—equals the foregone interest on the Z bond. Restructuring the last sequential as a Z bond shortens the average lives of all tranches in the structure, as deferred interest accelerates the amortization of the shorter classes (**Figure 47**).

Moreover, the average lives of all tranches are less volatile than structures without a Z bond (**Figure 48**). The placement of the Z at the back of the structure often protects it for a long time from prepaying too quickly. At the same time, its interest being directed to the earlier tranches keeps them from feeling the full effect of rate moves.

Figure 47: Weighted-average lives on a sequential structure with the LCF structured as a simple sequential or Z bond at 150 PSA

Tranche	Size	Average Life (years)		Effect	% Change
		D = Seq.	D = Z		
A	40,000,000	3.3	2.7	Shortened	-17%
B	25,000,000	8.1	6.1	Shortened	-24%
C	10,000,000	11.9	8.3	Shortened	-31%
D	25,000,000	19.3	15.9	Shortened	-18%
Pass-Through	100,000,000	9.3	9.3		

Source: J.P. Morgan

Figure 48: Percent change in the weighted average life of each sequential class when the speed increases from 150 PSA to 300 PSA

Tranche	Average Life (years)	
	D = Seq.	D = Z
A	-32%	-27%
B	-40%	-33%
C	-42%	-33%
D	-38%	-34%

Source: J.P. Morgan

Restructuring the last sequential as a Z bond also shortens the duration of earlier tranches but lengthens the duration of the Z due to lockout and its rising principal

balance. Because a Z bond has zero cash flow in the accrual phase, the cash flows are weighted towards the end of the Z bond's life, extending its duration. In contrast, the duration of earlier tranches is reduced. A Z bond can be converted back into a pass-through when the rest of the structure has paid off.

The price of a Z is highly sensitive to changes in interest rates because of its longer duration. Z bonds therefore typically offer higher yields than comparable sequentials or zero-coupon Treasuries. They also have more concentrated partial durations.

Z bonds can also be structured off PACs or support classes. In each case, the principal balance of the Z rises while its coupon gets diverted to other parts of the structure. Once the other bonds pay off, the Z begins to get principal and interest.

Z bonds offer the mortgage market's longest duration. The rising principal balance and the absence for years of any coupon from the Z leads to durations that can rival that of a 30-year Treasury bond. The structure often appeals to portfolios looking for substitutes to long Treasuries, agencies, or corporate bonds.

XII. VADMs

Very Accurately Determined Maturity classes or VADMs address an entirely different concern from Zs: extension risk. VADMs can provide a guaranteed final date for receipt of MBS principal and a maximum weighted average life even if prepayments drop to zero. For investors that need certainty around final return of principal, this structure can have significant value. VADMs show up often and the nature of their structure means they are typically a small portion of the overall deal.

Figure 49: VADM essentials

Key Features of VADMs	Easily determined maximum average life
	No risk to structure from interest rate whipsaw
Common Uses	Limit extension risk

Source: J.P. Morgan

A Very Accurately Determined Maturity Bond or VADM is a CMO class designed to provide certain protection against extension risk. Its total cash flows are derived from the foregone interest on a Z bond. To ensure that VADMs are protected against extension, the last cash flow of the VADM is determined by sizing the Z and VADM bonds at 0 PSA. At that speed, the last cash flow of the VADM is determined by the timing and size of the last accrual from the Z.

Because interest accruing from the Z tranche is buffered from short-term changes in prepayments, the VADM can sufficiently pay down its scheduled principal and interest should prepayment speeds fall. The average lives of VADMs can remain stable within a range of prepayment speeds. Typically, VADMs are structured to have a specific WAL at 0 PSA, and the exact range of speeds up to where that WAL is maintained depends on the amount of front cashflow. VADMs consequently have a fixed maturity until prepayment rates rise and cause the last sequential-pay tranche to pay off. At this point, the principal balance of the Z bond stops accruing and the VADM begins receiving principal earlier than it would if prepayment speeds were slower. **Figure 50** shows how the weighted-average life of a VADM will remain constant at slower prepayment rates (in this example at 5.1 WAL) and will decrease at prepayment rates above the pricing speed.

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Figure 50: A Z/VADM structure at different prepayment speeds

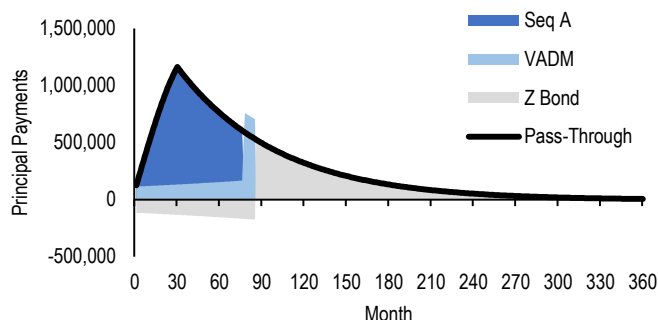
Shaded highlighting represents protection against extension risk

Speed	Seq A	VADM	Z
0 PSA	11.9	5.1	24.7
50 PSA	8.1	5.1	21.9
100 PSA	6.0	5.1	18.9
150 PSA	4.7	5.1	16.3
200 PSA	3.9	5.0	14.1
250 PSA	3.4	4.7	12.2
300 PSA	3.0	4.4	10.8
500 PSA	2.2	3.4	7.2

Source: J.P. Morgan

Figure 51: Principal cashflows at a faster prepayment speed than the upper bond of the VADM

Z/VADM Tranche Payments at 225 PSA



Source: J.P. Morgan

At higher prepayment speeds, earlier classes mature faster. Should these bonds mature before the VADM pays down completely, the Z bond's amortization may be deferred until the VADM is gone (**Figure 51**). All principal cash flows from the underlying collateral, as well as accruing interest on the Z bond, are directed to paying down the VADM. Once the VADM matures, the Z bond begins amortizing its accreted balance.

VADMs fit extremely well in portfolios that want protection against extension risk and that want to avoid the kinds of structural changes found in PACs as interest rates vary. The bands on PACs can change over time, and a period of fast prepayments followed by slow prepayments could lead to unexpected extension. VADMs do not have whipsaw risk.

XIII. GMCs

Note: Republished from our [June 5, 2026](#) weekly publication

The newest frontier in CMO structuring is the “guaranteed maturity class” (GMC), also referred to as mandatory redemption (MR); these are bonds with a hard final payment date guaranteed by the GSEs who will redeem outstanding principal at that time at par. Both Fannie and Freddie make clear in the offering documents that this guarantee is as rock solid as the implicit guarantee that bondholders won't take any losses on GSE-sponsored MBS. The GSEs, primarily Freddie, have issued almost \$12bn in GMCs during the past year and almost \$7bn over the past three months (**Figure 52**). Generally we think it makes the most sense to compare these structures to other spread products with hard finals, rather than other MBS products, as they will best fit into portfolios constrained to shorter bond terms.

As shown in the table, there are (at this point) two main flavors of GMCs depending on the collateral. GMCs backed by ARMs have followed a pattern of cutting off the tail, that is, setting the maturity date right before the first reset. This is why they primarily fall into the 5/7/10yr term buckets, or shorter if using slightly seasoned collateral. For buyers of these bonds, running ARMs to balloon at reset is no longer just a helpful convention but the reality of these cashflows. On the fixed-rate side, the choice of cutoff is more discretionary as there is no natural balloon point, but that's a concern for the GSEs redeeming the cashflows, as investors will still just see the normal cashflows on the PTs up until the final payment date. To date, 1yr and 5yr have been the most

common terms for GMCs off of 30yr collateral. The underlying bonds have been very vanilla up to this point: almost entirely generic conventional 5.5s (and a few 6s) on the fixed-rate side, and on the ARM side, mostly new issue bonds with a fixed-rate term matching the mandatory redemption option (mostly 7/6m).

Figure 52: GMC issuance picked up in earnest late in 2025 with structures backed by ARMs, and has been even more robust the past few months with 30yr PT collateral

Issuance of GMCs by collateral type (ARM or 30yr PT) and month, \$mm

Issuance \$mm	Final Maturity Term (rounded up, years)											Total
	ARM				Fixed-Rate							
Issuance month	5	6	7	10	1	2	3	5	7	10	11	
Jun-25										100		100
Jul-25										120		120
Aug-25												
Sep-25										100		100
Oct-25												
Nov-25	121	383	805							100		1,408
Dec-25	140	91	831							120		1,182
Jan-26	112		1,179	112						100		1,502
Feb-26			220					300	100			620
Mar-26	100		203		1,210			550			100	2,063
Apr-26			339		950		200	700		200		2,389
May-26			152	68	975	100	175	400	100	411		2,382
Total	473	474	3,729	180	3,135	100	375	1,950	200	1,251	100	11,868

Source: J.P. Morgan, Fannie Mae, Freddie Mac

To look at GMC valuations, we chose representative bonds issued in May across hard finals and used rough estimates of pricing spreads. One thing to keep in mind is that the deals with fixed-rate collateral are typically structured as modifiable and combinable securities (MACS) with a range of fixed coupon options; the most likely to be issued are those just under par and those are the bonds we chose to run. **Figure 53** shows the key valuation measures. As one would expect, the convexity/option cost is significantly lower on these bonds versus a typical par priced 30yr MBS, particularly for a 1yr hard final.

Figure 53: Analytics on a range of GMC maturities using recently issued bonds (as of 6/1/2026)

Hard Final	Bond	Coupon	Price	I Spread	OAS	OAD	OAC	Option Cost
1yr	FHS 451 GJ	4.25	99.9	25	19	0.9	-0.2	2
3yr	FHS 455 AH	4.50	99.9	40	25	2.0	-1.0	14
5yr	FHS 458 AB	4.50	99.3	55	26	2.7	-1.4	25
7yr	FHR 5666 GB	4.50	98.8	60	27	3.1	-1.1	25

Source: J.P. Morgan, Freddie Mac

XIV. Derivatives

The term “derivatives”, in the context of agency MBS, refers to structures that reference an underlying pool or pool of mortgages and receives only interest payments (no principal). They can come in either interest only (IO) form, and receive a fixed coupon, or as inverse interest only (IIO), and get paid a floating coupon that moves in the opposite direction of a benchmark. An IO can be made from interest stripped off most of the CMO classes described above, with PACs and pass-throughs being the most

common in recent years. In the simplest case, a pass-through can be split into just a lower coupon principal-bearing bond and an IO that receives the rest of the coupon. IOs bring along the same range of structural features as the parent class, including lockout from amortization and prepayments, different principal reference windows, average lives, and prepayment protection, leading to a diverse range of possible bonds. The other method of creating IOs involves the securitization of excess servicing (XS IOs) from borrower interest payments beyond what is needed for the MBS coupon, guarantee fee, and required 25bps retained servicing. IIOs are most often created to go along with a principal-bearing floater, as described above.

The interest-only nature of derivatives leads them to have greater exposure to prepayments. When thinking through their exposure to rates, it's important to consider separately the impact of prepayments (and changes to their cashflow, thought of as current coupon duration) as well as the impact of the rate used for discounting future cash flows. When these rates move in tandem, in a parallel curve scenario, an IO will typically see the change in cash flows outweigh the discounting impact, leading to its negative duration. More broadly, both IOs and IIOs having exposure to the current coupon (and mortgage rates), increasing in price when spreads widen, giving investors another tool for managing their overall mortgage exposure. This also leads them to have a typically positive price reaction to increases in vol (unlike PTs), known as “dirty vega”, since the impact of a vol increase on the current coupon outweighs the pure price impact of more dispersed OAS paths in the model. We discussed this phenomenon in our [4/1/2022 commentary](#).

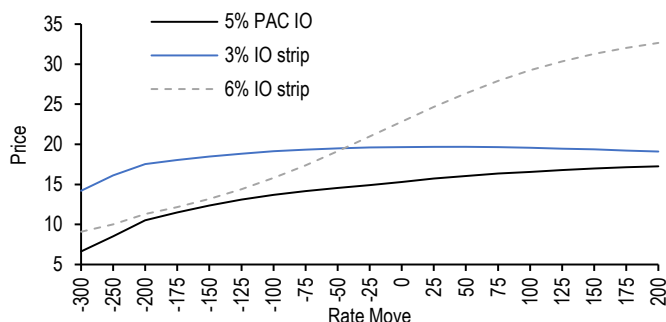
IOs

IOs, which only receive fixed interest payments, see their price pull to zero rather than the par typical of principal bearing bonds. As a result, the longer the underlying collateral stays outstanding, the better, leading to prices that rise along with interest rates. This negative duration feature is a key benefit to IOs and allows investors to shorten a portfolio's duration in an efficient manner. The heightened sensitivity of IOs to prepayments also means that a careful consideration of collateral and structure is necessary to manage their negative duration and convexity.

Model analytics provide a more tangible view into typical IO characteristics, as we show in the charts below using even-OAS model runs. **Figure 54** compares the price of three IOs in different rate moves at a prevailing mortgage rate of 6.5% in early 2023: 1) a slightly in-the-money 6% IO strip off new production 6s, 2) a very out-of-the-money 3% IO stripped from new production 3s, 3) an IO from a front-pay PAC made from generic 5s, in a structure that includes a Z bond. The 6 IO shows the reactivity of price to rates, especially close to at the money, while the 3 IO is so in the money that it has minimal negative duration. The IO from a PAC structure is less reactive in price overall than either of the other examples as one would expect with a shorter principal window, though of course that can vary depending on both the collateral and structure.

Figure 54: Using a few different IOs to demonstrate the impact of coupon and structure on prices

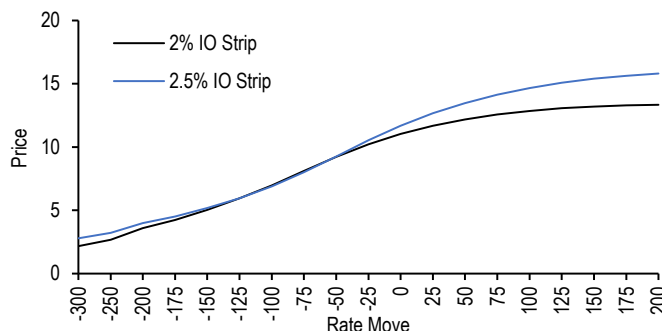
Model price for a few different hypothetical IO structures in different rate scenarios, 1/3/2023



Source: J.P. Morgan

Figure 55: Typical price profile of near at-the-money IOs in the more normal rate environment of early 2022

Model price for a few different hypothetical IO structures in different rate scenarios, 1/4/2022

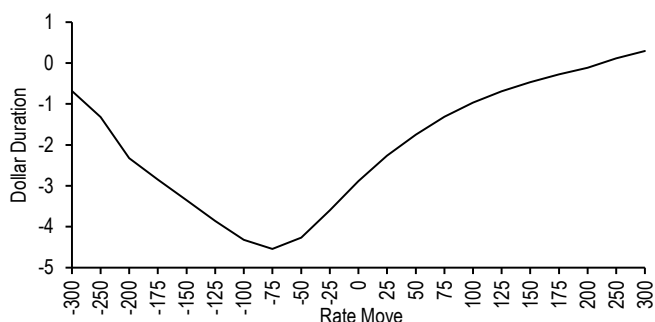


Source: J.P. Morgan

Figure 55 shows the price profile of close to at-the-money IOs (2% and 2.5% collateral) in the more normal rate environment of early 2022. Focusing on the at-the-money IO, **Figure 56** and **Figure 57** show the dollar duration (change in price for a 100bp move in rates) and dollar convexity (change in dollar duration for a 100bp move in rates). The duration is negative across all rate scenarios, and is most negative **around 50bp in the money**.

Figure 56: IO duration reaches its most negative levels as the bond prepayments pick up in a rally ...

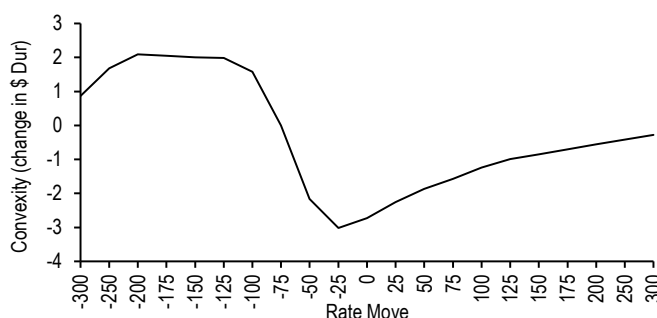
At-the-money IO dollar duration (price change for 100bp move in rates) across a range of rate scenarios (using 2% IO on 1/4/22)



Source: J.P. Morgan

Figure 57: ... and convexity is lowest near at-the-money, and can even become positive into a large enough rally where changes in rate have little impact on prepayments

At-the-money IO dollar convexity (dollar duration change for 100bp move in rates) across a range of rate scenarios (using 2% IO on 1/4/22)



Source: J.P. Morgan

In terms of structure, an IO created from a short sequential class with an open principal window, for example, can be much more sensitive to changes in interest rates than an IO stripped down from unstructured pass-throughs (“Strip IO”). The short sequential IO is immediately exposed to prepayment risk and could be extinguished in a single prepayment wave, while the strip IO might remain outstanding much longer. An IO stripped off a stable 5-year PAC currently locked out from prepayments, by contrast, would typically be much less sensitive than a strip IO. The PAC structure and the lockout protects the IO over a range of interest rates from erosion of the underlying principal. An investor wary of a short-term drop in interest rates but looking for a long-term rise might look for structured IO with a few years of principal lockout—protection against an immediate dip in rates. Unlike the underlying structure, a whipsaw in rates

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could be beneficial for the PAC IO, since it possibly carried just enough projection on the cashflow and is helped by the extension on the other side.

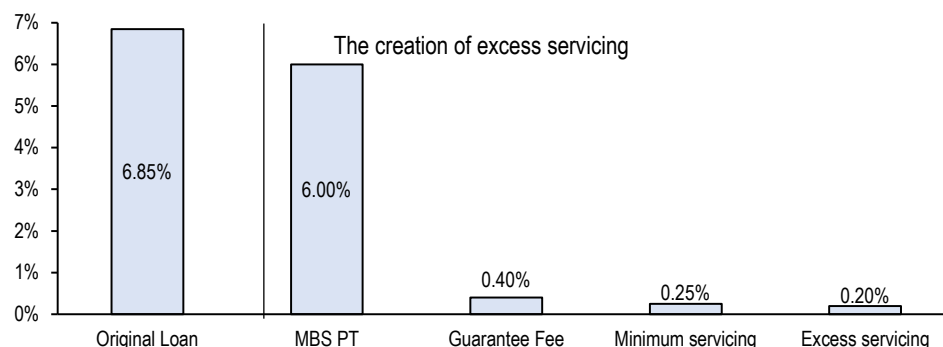
When valuing IOs, the simplest approach is to estimate static prepayments on the underlying collateral and see the resulting yield. This approach has more relevance to IOs far away from the money, but still suffers from not building in the inherent optionality of MBS. Model-based OASs are a more common point of comparison. IOs' sensitivity to prepayments extends to OAS, increasing the impact of the underlying prepayment model. The market closely tracks the performance of model versus actual speeds and the ramifications of model updates.

In terms of their use, the negative duration of the IO asset along with an attractive yield makes it a good tool for shortening a portfolios overall duration. As opposed to selling Treasuries, IOs can bolster yield and enhance carry. A popular IO trade is to go long both a mortgage pool (or TBA) and IO to achieve a 0 duration profile with significant carry relative to MBS hedged with Treasuries. The MBS + IO position is also more hedged to changes in overall mortgage spreads as IOs benefit from spread widening (and slower prepayments), as we mentioned in the introduction on current coupon duration in derivatives. The trade-off for these features is additional negative convexity from the MBS and IO positions. This can be mitigated (or planned for) by the choice of collateral on both the mortgage pool and IO.

Excess IOs (XS IOs)

The distinction between the structured IOs above and XS IOs is purely in their creation process, and the principles discussed above for valuations and usage are largely the same between both. In this section, we'll focus on what makes XS IOs different and the nuances of different deal structure. First, it's helpful to understand the building blocks for XS deals. The "XS" or "excess yield" is essentially the servicing leftover after a mortgage is pooled, beyond the mortgage coupon, agency "base" guarantee fee without buy-ups, the minimum servicing fee, and any additional servicing. Consider a simplified example where a servicer is pooling a loan with a 6.85% rate into a 6% pass-through (**Figure 58**). After the pass-through gets paid, 85bp of interest remains, of which 25bp must be retained by the servicer, and roughly 40bp is paid to the GSE for its guarantee. The 20bp = 85bp - 40bp - 25bp leftover is then paid to the servicer and known as excess yield.

Figure 58: The creation of excess servicing when a loan is packaged into a pass-through



Source: J.P. Morgan

Over time, servicers can amass substantial positions in “excess” servicing that they aren’t required to hold. Additionally, the GSEs allow for servicers to “buy-up” their guarantee fee (pay more than the required g-fee in exchange for upfront cash), leading to Fannie Mae and Freddie Mac themselves holding onto excess yield. This leads to the two types of XS IO deals: 1) XS IO from a particular servicer on loans where they retain the servicing and 2) IO from the GSEs using the buy-ups. Both deals employ similar structures, and the main distinction will be in the actual servicer on the loans, either the single large servicer or a mixture based on the GSEs buy-up holdings.

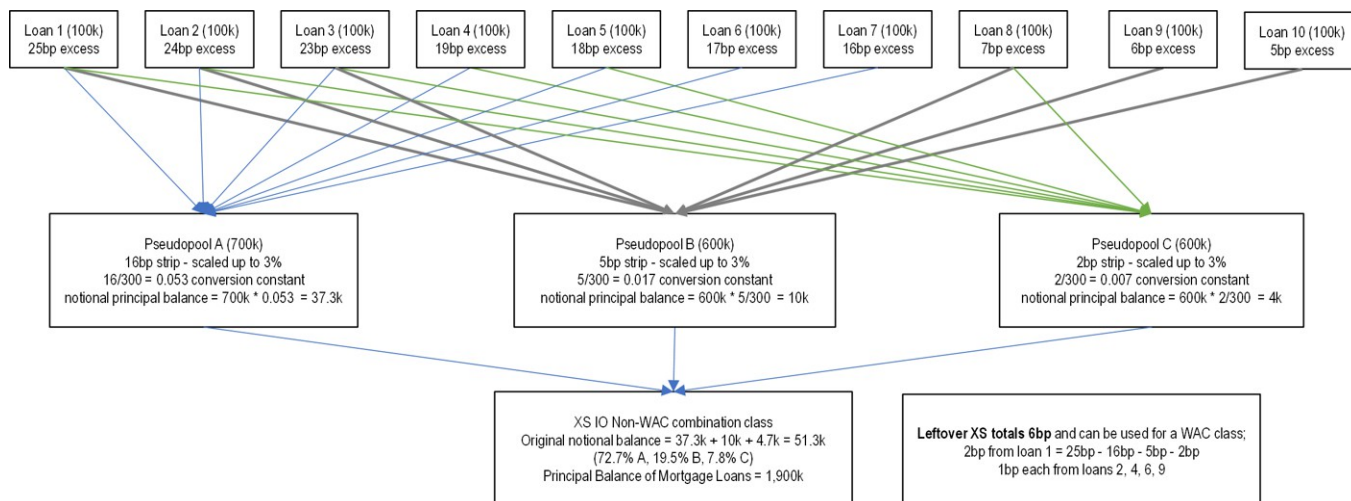
Servicers have a variety of reasons for why they might choose to sell XS IOs. As opposed to an MSR transaction, they maintain the borrower relationship, allowing for recapture possibilities on refi or cross-selling. They can use an XS deal to raise cash and/or shed MSR for capital reasons. Mults are often higher than the raw MSR asset for liquidity and financing reasons.

Moving onto structure, these deals are packaged via the GSEs, and the interest payments are guaranteed. When the deal is created, the interest going into it is no longer part of the servicing strip in the case of a servicer securitizing IO. This means that the servicer can still transfer their remaining servicing per the normal GSE guidelines. Crucially, a loan’s inclusion in the XS IO deal is directly linked to its presence in the MBS pass-through, so the same principles related to prepayments and defaults can be carried over. This language is typical: “As long as a Mortgage remains in the applicable UMBS, MBS or Gold PC pool, Freddie Mac will pay the Excess Yield attributable to that Mortgage to the Holders of the applicable Class or Classes of SCs.” Where things get interesting is how the exactly the XS from each loan gets counted for in the bond classes. The main distinction is between non-WAC classes (the vast majority of the deal) and WAC classes.

Non-WAC class (Figure 59). These classes are structured to pay interest at the same rate as the underlying pass-through by scaling the notional balance. To accomplish this, loans are grouped by coupon first, and by term, seasoning, and potentially spec characteristics for best execution. One subtlety in the construction is that within any grouping, loans will have a range of excess yield available. To avoid a proliferation of bonds, loans can contribute different amounts of coupon to the overall pool.

For calculation purposes, the loans are first aggregated into “pseudopools” where each has the same excess yield contribution, and then any number of these pseudopools are combined to form the top-level combined non-WAC class. When constructing the pseudopools, the structurer’s objective is to maximize the amount of excess yield they contain (for a given number of pseudopools) to get the most coupon directed into the non-WAC class (with the rest going to the WAC class). The example below shows one way of doing that for a given set of 10 loans. Conceptually this is very similar to how other structured CMOs can be built from fractions of pools, and the prepayment math works in a similar manner. The overall combined non-wac class collateral statistics will be proportional to what share of its balance is contributed from each pseudopool.

Figure 59: The path from excess IO on a set of loans to a single XS IO non-WAC class in an optimal manner with three pools (more pools can reduce the leftover XS by a greater amount)

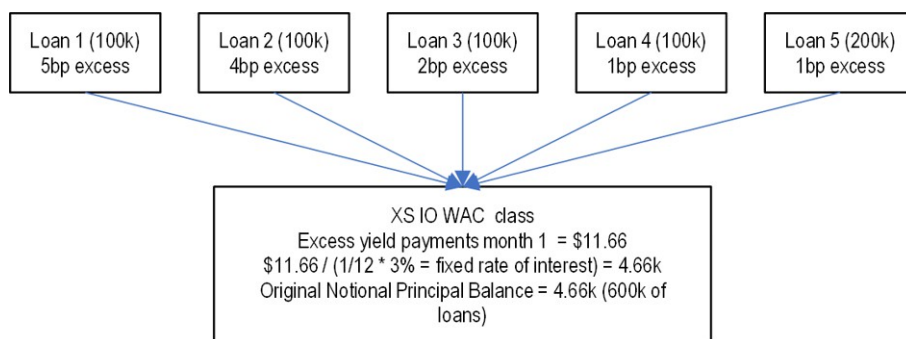


Source: J.P. Morgan

WAC-class (Figure 60). XS IOs are structured to get as much interest as possible into the non-WAC classes, but even so, there can be a small amount of leftover WAC in the various collateral cuts. This additional coupon is placed together in the WAC-classes. For both Fannie Mae and Freddie Mac WAC-classes, the first month notional is scaled to correspond to the coupon of the underlying pass-through given the amount of interest coming through all the loans. For example, the WAC-class might be paying a weighted average of 5bp on an underlying \$100mm of FNCL 3s. The interest payment is $\$100mm \times 5bp \times (1/12) = \4.167 on a given month, which is equivalent to the 3% payment on \$1.666mm, which is the notional value that will be shown for the WAC-class.

However, the variable contributions from different loans make the accounting messier as time goes on. The agencies have a different approach to marking these tranches each month; Fannie Mae will resize the notional based on the actual interest received, while Freddie Mac allows the WAC to drift depending on which loans pay off. In either case, the variable interest contributions by loan make it difficult to exactly track the prepayments on the collateral.

Figure 60: The initial structure of a WAC class



Source: J.P. Morgan

Inverse IOs (IIOs)

IIOs are structured CMO tranches that offer a monthly floating coupon based on a strike rate minus a reference rate, which was 1-month LIBOR historically and for newer production is 30-day average SOFR. The IO part of the name comes from the fact that the bond receives no principal payments (although notional pays down along with collateral), while the inverse part of the name is due to the bond coupon falling as interest rates increase, in contrast to traditional floaters.

Inverse IOs are created as a by-product of constructing floaters from fixed-rate collateral. The floater receives the principal payments from collateral, along with a floating coupon. The interest not paid to the floater is used to create an inverse IO. The level of the reference rate at which the floater reaches its cap sets the floor of the inverse IO and is referred to as the strike. When the floater reaches its maximum coupon, the inverse IO falls to its minimum coupon. For example, creating a 7% cap floater receiving the reference rate (R) + 50bps would result in an inverse IO receiving 6.5% - R. If the reference rate climbs to 6.5%, the floater gets a 7% coupon, while the inverse IO receives no coupon. Conversely, if R falls to zero, the floater will earn 50bps of coupon, while the inverse IO picks up 6.5%.

Inverse IOs can also be created with added coupon leverage. By reducing the notional balance, a CMO desk can create inverse IOs with a greater coupon. The dollar amount of coupon paid to the tranche remains constant; however, the smaller amount of notional may allow levered investors to obtain higher returns due to a potentially lower margin requirement on the smaller notional. For example, rather than creating the 6.5% - R security above, a desk can create a 13% - 2 x R inverse IO with half the balance of the original security. This security provides investors with a coupon of up to 13% of notional balance, helping meet high yield targets.

Inverse IOs are attractive to levered investors with a high return target. After duration and convexity risks, inverse IOs often trade with a wide OAS. This OAS concession is a result of some liquidity concessions and the heightened prepayment and rate sensitivity relative to a floater or fixed-rate collateral. Fund managers who can manage the risks embedded in an inverse IO can pick up spread in the product.

Inverse IOs (IIO) are subject to prepayment risk as well, with fast prepayments decreasing the value of the security and slow prepayments leading to more cash flows. However, the IIOs also have coupon/cap risk. The coupon on an inverse IO decreases in a selloff as the index rate approaches the strike. This effect has the potential to dominate any increase due to slower prepayments, so that inverse IOs can have positive duration, depending on the collateral characteristics and structure.

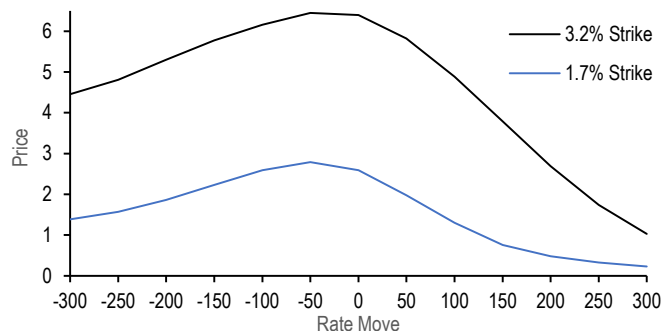
Inverse IOs have negative convexity, but less so than traditional IOs. This is due to the nature of the coupon. As rates rally, faster prepayments shorten the window of cash flows, which reduces bond value, but the coupon increases, which increases bond value. At low enough rates, the coupon caps out, which leads to the duration turning negative due to the shorter cashflows.

Figure 61 and **Figure 62** demonstrate these principles with a couple of IIOs. On the price chart, both the higher and lower strike IIOs increase in value into an initial rally, before slowly decreasing into a deeper rally when the coupon maxes out. They lose value into selloffs as the coupon disappears. Level-wise, the higher strike prices a fair amount higher for its greater coupon, but this gap can vary depending on the actual structure and collateral. The chart on the right uses the higher strike IIO to demonstrate

typical IIO greeks. Convexity is most negative around at-the-money and can eventually turn positive into a very deep rally or sell-off. Duration is positive in the middle rate region and can go negative into a sell-off. Then to wrap up our look at derivative valuations, **Figure 63** compares the price shifts on an IO and IIO with the same collateral (G2 3s).

Figure 61: The price profile of IIOs with different strikes

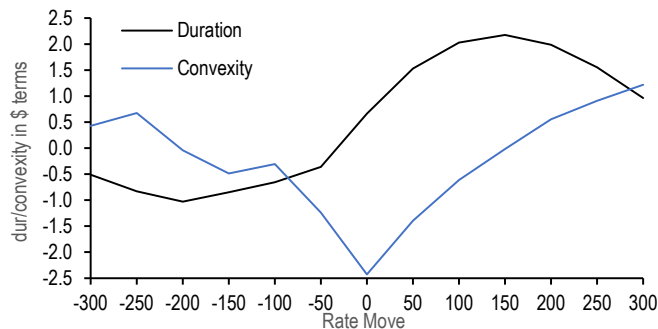
Price in different rate scenarios for two IIOs based on G2 MA pools: 1.7% strike GNR 2021-193 CS (G2 2.5s) and 3.2% strike GNR 2021-205 SL (G2 3s) on 1/3/22



Source: J.P. Morgan

Figure 62: Duration and convexity profile of an IIO

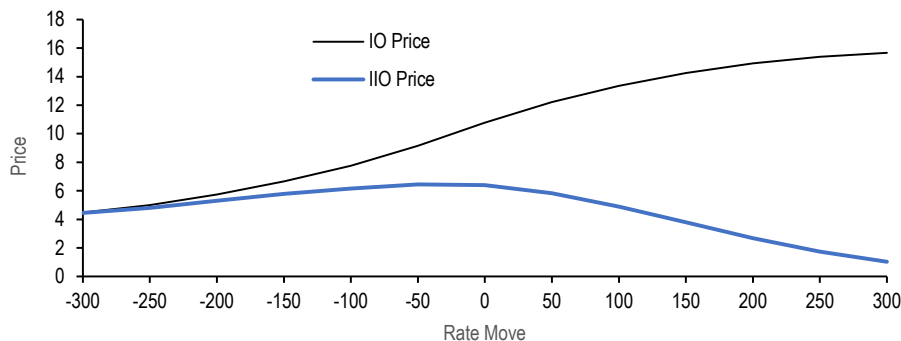
Dollar duration and convexity on GNR 2021-205 SL in different rate scenarios on 1/3/22



Source: J.P. Morgan

Figure 63: An example of IO and IIO price reactivity on the same collateral

IO versus IIO prices in rate moves using two bonds with the same collateral (and similar coupon at SOFR = 0%), GNR 2021-205 IL and GNR 2021-205 SL on 1/3/22



Source: J.P. Morgan

The relationship among rates, prepayments, and coupon paints an interesting partial duration profile in inverse IOs. Increases in the short rate will decrease bond coupon but leave prepayments largely unaffected. Hence, 2-year and 5-year partials are positive. Increases in longer rates will slow borrower prepayments but are less meaningful for bond coupon, hence 10-year partials are negative. Owning an inverse IO without hedging the curve is effectively holding a steepener. For this reason, investors in inverse IOs must be particularly concerned about flattening curve scenarios.

Unlike IOs, investors can hedge the positive duration of inverse IOs by selling pass-through securities, or they can buy TBA and sell Treasuries to hedge both the current coupon and interest rate duration. Swaptions and Treasury options can also be included to express different interest rate views.

For more details on the current coupon hedging needs associated with the IIO, IO, and MSR, please see our pieces from [5/8/2026](#) and [5/15/2026](#).

XV. Data Appendix

Our primary report for CMO is the “CMO & IO Issuance Report” which covers monthly issuance trends and collateral details, available through our [JPMM page](#) on agency MBS.

For real-time analytics on a given CMO using the JP Morgan prepayment model, clients can use the [Multi Security Calculator](#) or the [OASis interface](#).

The rest of our data for CMOs is stored in [DataQuery](#).

The CMO information is under the following pathway: US Fixed Income/ MBS / Agency MBS / CMOs (**Figure 44**). The subcategories are as follows:

- Indicative Pricing
 - PAC/SEQ: Weekly indicative spread levels for these structures split into average life ranges (**Figure 45**). Information is stored as far back as 2002. These are meant to represent current issue at the time so the coupon/collateral assumptions change, and the latest populated term (10yr or 20yr) should be assumed to be the last cash flow
 - Floater: A few collateral / cap combinations possible, depending on their relevance for actual issuance. Discount margin is in an indicative level marked weekly
- Indices: For principal-bearing classes, analytics on bonds matching specific structure/collateral/year/bond coupon, based on PricingDirect prices. We created the indices by looking at commonly issued structures each year and grouping by coupon and agency.
 - Floater/IIO Indices (**Figure 66** and **Figure 67**): We have introduced new floater/ IIO indices towards the beginning of each year over the past several years, generally grouping by agency, cap (for floaters), strike, generic/loan bal, collateral coupon, and issue year (see our [2023 and prior](#), [2024](#), and [2025](#) updates for more details on methodology).
 - IO Indices (**Figure 68**): In the spirit of IOS and trust IOs, we created our own IO indices to give an overall sense of valuations for new structured IO. We apply various rules to bond selection (only strips, newer issue) and look for combinations of agency/spec type/coupon that have sufficient balance for tracking. Pricing for the indices is aggregated from the bond-level courtesy of PricingDirect. For more details on our IO indices, see these pieces on the 2021 edition ([initial document](#), [update](#))
- Legacy Indices (IOS-POS-MBX, **Figure 69**): Analytics on these synthetic structures during the periods when they were relevant, useful for doing historical valuation work

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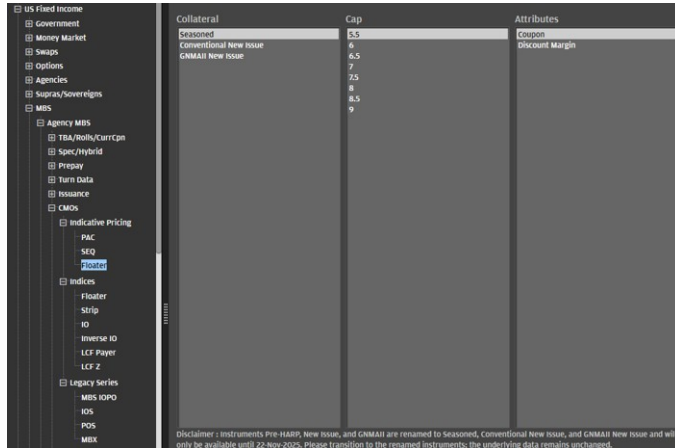
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21 July 2026

Figure 64: Selection options for the floater indicative pricing tab



Source: J.P. Morgan

Figure 65: Weekly CMO spreads for PACs and sequentials as shown on our daily packet

CMO Spreads (As of 07/10/2026)	PAC1 (5.0x5.0)	5D	SEQ (5.0x5.0)	5D
2 Yr	66	4	70	4
3 Yr	74	4	78	4
5 Yr	90	4	94	4
LCF	102	4	109	4

Source: J.P. Morgan

Figure 66: Selection options for the Floater indices tab

Indices	Attribute	PP Model#
FLT.GN.750.2023	Price	Default Model(2025)
FLT.GN.800.2023	DM (Forward)	2025 model
FLT.CONV.GEN.650.555.CC60.2024	DM (Static)	2024 model
FLT.CONV.GEN.650.555.CC65.2024	Labor OAS	2023 model
FLT.CONV.GEN.700.560.CC60.2024	Labor Zspread	2022v2 model
FLT.CONV.GEN.700.560.CC65.2024	Labor OA Duration	2022 model
FLT.CONV.GEN.750.565.CC65.2024	Labor OA Convexity	2021v2 model
FLT.CONV.GEN.800.570.CC65.2024	Labor OAS Spread Duration	2021 model
FLT.GN.GEN.600.545.CC60.2024	Tsy OAS	
FLT.GN.GEN.650.555.CC65.2024	Tsy Zspread	
FLT.GN.GEN.650.555.CC60.2024	Tsy OA Duration	
FLT.GN.LNBAL.650.555.CC60.2024	Tsy OA Convexity	
FLT.GN.LNBAL.650.555.CC65.2024	Tsy OAS Spread Duration	
FLT.GN.GEN.700.560.CC65.2024	Avg Life	
FLT.GN.GEN.700.560.CC70.2024	Yield	
FLT.GN.LNBAL.750.560.CC65.2024	Nominal Spread	
FLT.CONV.GEN.600.545.2025	CPR Hist 1M	
FLT.CONV.GEN.650.555.2025	CPR Hist 3M	
FLT.CONV.LNBAL.650.555.2025	CPR Hist 6M	
FLT.CONV.LNBAL.650.550.2025	CPR Proj 1M	
FLT.CONV.GEN.650.550.2025	CPR Proj 3M	
FLT.CONV.GEN.700.560.2025	CPR Proj 6M	
FLT.CONV.LNBAL.700.560.2025	CPR Proj 1Y	
FLT.CONV.GEN.800.570.2025	CPR Proj Life	
FLT.GN.GEN.600.545.2025	SOFR OAS	
FLT.GN.LNBAL.650.550.2025	SOFR OA Duration	
FLT.GN.GEN.650.550.2025	SOFR OA Convexity	
FLT.GN.LNBAL.650.555.2025	SOFR OA Spread Duration	
FLT.GN.GEN.650.555.2025	SOFR Zspread	
FLT.GN.LNBAL.700.560.2025		
FLT.GN.GEN.700.560.2025		
FLT.GN.LNBAL.750.565.2025		
FLT.GN.GEN.800.570.2025		

Disclaimer: Attributes evaluated using Libor were discontinued in Jan, 2024.

Source: J.P. Morgan

Figure 67: Selection options for the IIO indices tab

Indices	Attribute	PP Model#
IIO.CONV.GENERIC.545.C60.2025	Price	Default Model(2025)
IIO.CONV.GENERIC.550.C60.2025	Labor OAS	2025 model
IIO.CONV.GENERIC.550.C65.2025	Labor Zspread	2024 model
IIO.CONV.GENERIC.555.C60.2025	Labor OA Duration	2023 model
IIO.CONV.GENERIC.570.C60.2025	Labor OA Convexity	2022v2 model
IIO.CONV.LOANBAL.550.C60.2025	Labor OAS Spread Duration	2022 model
IIO.CONV.LOANBAL.555.C60.2025	Tsy OAS	2021v2 model
IIO.GN.GENERIC.545.C60.2025	Tsy Zspread	
IIO.GN.GENERIC.550.C60.2025	Tsy OA Duration	
IIO.GN.GENERIC.550.C65.2025	Tsy OA Convexity	
IIO.GN.GENERIC.555.C60.2025	Tsy OAS Spread Duration	
IIO.GN.LOANBAL.550.C60.2025	Avg Life	
IIO.GN.LOANBAL.550.C65.2025	Yield	
IIO.CONV.LOANBAL.555.C60.2025	Nominal Spread	
IIO.CONV.GENERIC.550.C60.2024	CPR Hist 1M	
IIO.CONV.GENERIC.550.C65.2024	CPR Hist 3M	
IIO.CONV.GENERIC.555.C60.2024	CPR Hist 6M	
IIO.CONV.GENERIC.555.C65.2024	CPR Proj 1M	
IIO.CONV.GENERIC.555.C70.2024	CPR Proj 3M	
IIO.CONV.LOANBAL.550.C65.2024	CPR Proj 6M	
IIO.CONV.LOANBAL.555.C60.2024	CPR Proj 1Y	
IIO.GN.GENERIC.550.C60.2024	CPR Proj Life	
IIO.GN.GENERIC.550.C65.2024	SOFR OAS	
IIO.GN.GENERIC.555.C60.2024	SOFR OA Duration	
IIO.GN.GENERIC.555.C70.2024	SOFR OA Convexity	
IIO.GN.LOANBAL.550.C60.2024	SOFR OA Spread Duration	
IIO.GN.LOANBAL.555.C60.2024	SOFR Zspread	
IIO.GN.LOANBAL.555.C70.2024		
IIO.CONV.GENERIC.550.C65.2023		
IIO.CONV.GENERIC.555.C65.2023		
IIO.CONV.GENERIC.560.C65.2023		

Disclaimer: Attributes evaluated using Libor were discontinued in Jan, 2024.

Source: J.P. Morgan

Figure 68: Selection options for the IO Indices tab

Indices	Attribute	PP Model#
IO.CONV.GENERIC.200.2020	Price	Default Model(2025)
IO.CONV.GENERIC.250.2020	Libor OAS	2025 model
IO.CONV.GENERIC.300.2020	Libor ZSpread	2024 model
IO.CONV.GENERIC.350.2020	Libor OA Duration	2023 model
IO.CONV.GENERIC.400.2020	Libor OA Convexity	2022v2 model
IO.CONV.JUMBO.200.2020	Libor OAS Spread Duration	2022 model
IO.CONV.JUMBO.250.2020	Tsy OAS	2021v2 model
IO.GN.GENERIC.200.2020	Tsy ZSpread	
IO.GN.GENERIC.250.2020	Tsy OA Duration	
IO.GN.GENERIC.300.2020	Tsy OA Convexity	
IO.GN.GENERIC.350.2020	Tsy OAS Spread Duration	
IO.GN.GENERIC.400.2020	Avg Life	
IO.GN.JUMBO.200.2020	Yield	
IO.GN.JUMBO.250.2020	Nominal Spread	
IO.CONV.GENERIC.200.2021	CPR Hist 1M	
IO.CONV.GENERIC.250.2021	CPR Hist 3M	
IO.CONV.GENERIC.300.2021	CPR Hist 6M	
IO.CONV.LOANBAL.200.2021	CPR Proj 1M	
IO.CONV.LOANBAL.250.2021	CPR Proj 3M	
IO.CONV.LOANBAL.300.2021	CPR Proj 6M	
IO.CONV.LOANBAL.350.2021	CPR Proj 1Y	
IO.CONV.NY.200.2021	CPR Proj Life	
IO.CONV.NY.250.2021	SOFR OAS	
IO.CONV.NY.300.2021	SOFR OA Duration	
IO.CONV.NY.350.2021	SOFR OA Convexity	
IO.CONV.JUMBO.200.2021	SOFR OA Spread Duration	
IO.CONV.JUMBO.250.2021	SOFR ZSpread	
IO.GN.GENERIC.200.2021	PDUR_2Y_TSY	
IO.GN.GENERIC.250.2021	PDUR_5Y_TSY	
IO.GN.GENERIC.300.2021	PDUR_10Y_TSY	
IO.GN.GENERIC.350.2021	PDUR_30Y_TSY	
IO.GN.LOANBAL.200.2021	VEGA	

Disclaimer : Attributes evaluated using Libor were discontinued in Jan, 2024.

Source: J.P. Morgan

Figure 69: Selection options for the IOS tab

MBS Agency Index IOS Constituents		
The dataset focuses on MBS Agency Index IOS Constituents, featuring 27 Deal Names and a variety of Attributes like Average Life, Coupon, and CPR metrics. It provides options for different PP Models from 2010 to 2020 and covers financial data in USD specific to the United States. The menu is constructed of 1 dimension(s): Deal Name (e.g. IOS.FN30.300.12(3.0), IOS.FN30.300.13(3.0), IOS.FN30.350.10(3.5),...). This dataset is produced by JP Morgan MBS Research Team. DataQuery API group id = FI.MBS_IOS_IOS		
Deal Name (27)	Attribute	PP Model#
IOS.FN30.300.12(3.0)	Average Life	2020C model
IOS.FN30.300.13(3.0)	Coupon (collateral IO)	2018 model
IOS.FN30.350.10(3.5)	CPR Hist 1m	2014 model (SMM3)
IOS.FN30.350.12(3.5)	CPR Hist 3m	2014 model
IOS.FN30.350.13(3.5)	CPR Hist 6m	2013 model
IOS.FN30.350.14(3.5)	CPR Proj 1m	2012 model
IOS.FN30.400.09(4.0)	CPR Proj 3m	2011 model
IOS.FN30.400.10(4.0)	CPR Proj 6m	Aug 2010 model
IOS.FN30.400.11(4.0)	CPR Proj 1y	2010 model
IOS.FN30.400.13(4.0)	CPR Proj Life	
IOS.FN30.400.14(4.0)	Factor	
IOS.FN30.450.09(4.5)	Hedge Ratio Mtge Curr Coupon	
IOS.FN30.450.10(4.5)	Hedge Ratio Tsy 10yr	
IOS.FN30.450.11(4.5)	Libor OAS	
IOS.FN30.500.03(5.0)	Libor ZSpread	
IOS.FN30.500.05(5.0)	Libor Option Cost	
IOS.FN30.500.08(5.0)	Libor OA Duration	
IOS.FN30.500.09(5.0)	Libor OA Convexity	
IOS.FN30.500.10(5.0)	Tsy ZSpread	
IOS.FN30.550.03(5.5)	Tsy OAS	
IOS.FN30.550.05(5.5)	Tsy Spread Dur	
IOS.FN30.550.08(5.5)	Vega	
IOS.FN30.600.08(6.0)	Mtge Spread Duration	
IOS.FN30.650.09(6.5)	Libor Spread Duration	
IOS.G230.400.10(4.0)	Prepay Duration	
IOS.G230.450.10(4.5)	Price	
IOS.G230.500.10(5.0)	Carry(bid)	
	Hedge Adj Carry(bid)	
	Carry(proj)	
	Hedge Adj Carry(proj)	
	PSA Projected Life	
	Settle Date	
	Yield	
	WAC(coll)	
	WALA(coll)	
	www.jpm.com	

Source: J.P. Morgan

XVI. (Legacy) Trust and Structured IO/PO

The IO/PO market falls into three areas:

- Trust IO/Pos (discontinued)
- Structured IO/POs (still active, described in earlier derivatives section), and
- IO/PO synthetics (discontinued, referred to as IOS)

Figure 70: Trust IO/PO essentials

Key Features of Interest-Only and Principal-Only Classes	Interest-Only (IO) Pays interest on underlying MBS but no principal Can be structured off of almost any underlying MBS Has negative duration—the price rising as rates rise, and falling as rates fall
	Principal-Only (PO) Pays principal on underlying MBS but no interest Can be structured off of almost any underlying MBS Often has a long duration and positive convexity
Common Uses	Interest-Only (IO) To shorten portfolio duration To add yield or carry express prepayment views
	Principal-Only (PO) To add portfolio duration or convexity express prepayment views

Source: J.P. Morgan

TRUST IO/POs

Trust IO/POs came from separating or stripping apart the interest and principal of pass-throughs. The pass-throughs go into a trust created solely to produce the new securities. The IO class receives all interest, and the PO class receives all principal. These are the pure building blocks of the Trust IO/PO market.

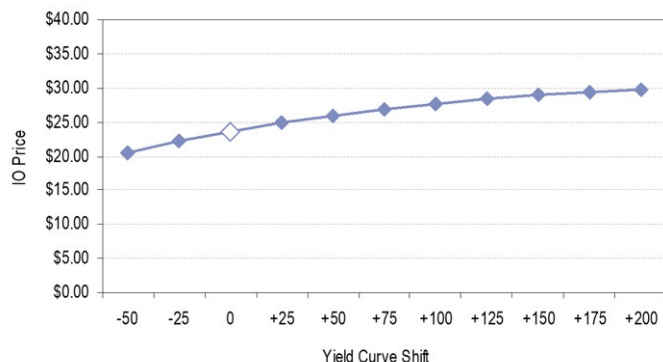
Historically, IO/PO trusts typically came together through the joint efforts of several dealers responding to investor demand. Billions of dollars used to go into new IO/PO trusts every year, although issuance collapsed after the rise of IOS (and its subsequent demise). Either Fannie Mae or Freddie Mac could set up the new trust, give it a number and coordinate the syndicate. Each participating dealer announced the amount of pass-throughs that the firm would contribute to the new trust. At settlement, the dealer paid a fee and exchanged that amount of pass-throughs for the same face amount of the new IO and PO. A dealer contributing \$100 million of pass-throughs to a new trust, for example, would receive \$100 million of the new IO and \$100 million of the new PO¹. The sum of the IO and PO cash flows matches those of the pass-throughs, making the transaction an even exchange. In fact, Fannie Mae and Freddie Mac would allow the dealer or any investor at any time to pay a fee and exchange a matched amount of IO and PO from the same trust for the same face amount of the underlying pass-throughs. Among other things, this prevented the combined price of the IO and PO from falling below the price of the corresponding TBA pass-through. If it did, the market would recombine the IO and PO and sell them at the higher TBA price.

Stripping the pass-throughs into IO and PO creates securities with prepayment and price profiles that differ dramatically from each other and from the parent pass-through. This

1. The trust also contains other classes besides the simple IO and PO. Synthetic Pass-through Securities or SPS combine IO and PO in different proportions, creating securities with coupons ranging from 0.50% to as much as twice the coupon of the underlying pass-throughs. These trade less frequently than the simple IO or PO. One advantage of SPS is to allow a portfolio to hold combinations of IO and PO without showing them separately on their books.

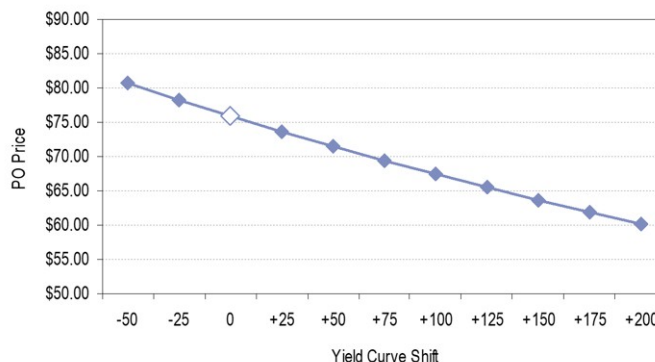
is part of the appeal of mortgage strips—the unique features of IO/PO make them useful in a much wider range of applications than the pass-through alone. A trust backed by pass-throughs priced at \$100-00 might produce an IO priced at \$25-00, for example, and a PO priced at \$75-00. As interest rates change, however, those prices begin to move in interesting ways, as we discussed in the section above for IOs.

Figure 71: IO price profile



Source: J.P. Morgan

Figure 72: PO price profile



Source: J.P. Morgan

POs benefit from falling interest rates, which drive prepayments up. Prepayments return principal at par, and, because PO investors usually buy the securities at deep discounts to par, the quick return of principal produces significant gains. POs consequently rise in price as interest rates fall. The opposite unfolds as rates move in the other direction. Higher rates put the brakes on prepayments, and the potential return of principal takes longer. PO prices consequently fall. This pattern usually gives a PO a long effective duration (**Figure 72**). And until the price of a PO approaches par—the price of the returned principal—it usually has positive convexity. In other words, a drop in interest rates drives up a PO price more than a comparable rise in rates drives it down. Because of this positive convexity, POs often trade at modest yields. POs become negatively convex if interest rates drop far enough to push prepayments to their maximum. At that point, any further drop in rates has limited impact on the PO cash flows. The PO price then has limited upside, and a much larger downside.

Historically, their simple structure and broad distribution gave IO/PO strips some of the best liquidity in structured MBS.

One of the most common uses of IOs is to help shorten a portfolio's effective duration. The change in the IO price helps offset the opposite changes in the price of other MBS or fixed-income securities. Because IOs also typically offer attractive yield, investors often see IOs as a more efficient hedge than taking a short position in Treasuries or in other securities. The IO may lack the liquidity of Treasury securities, but it compensates by actually adding to the yield of the portfolio. For portfolios that cannot sell securities short or that do not want to sell any existing holdings, IOs may be one of the only ways to shorten portfolio duration.

POs can help add duration, convexity, and prepayment protection to portfolios. MBS portfolios often include prepayment risk in various forms, and POs can partially offset the impact of rising prepayments. For portfolios that include IOs or MBS purchased at a premium to par, the offsetting benefits of POs can be critical if interest rates drop and send prepayments higher.

IO/PO SYNTHETICS (IOS)

The introduction of swap agreements in IO/PO market changed the way investors gained or shed exposure to IO/PO securities. IO and PO synthetic instruments offered increased liquidity in the market, simplicity in analysis, and leverage benefits. Unfortunately, the product, which started in earnest in 2011 with indices based on 2010 collateral, fizzled out by the start of 2017 as trading declined and no new indices were created, discussed in our write-up on 2/5/2016.

In an IO/PO synthetic contract, counterparties gained or shed exposure to the market by means of a total return swap. The long investor made a monthly financing payment to the short investor, usually at a rate of 1-month LIBOR on a previously agreed upon notional balance and received from the short investor a coupon whose amount referred to a particular cohort of IO/PO cash bonds. Like holding a cash IO/PO, the notional balance amortized according to the amortization schedule of the reference pools. Thus, the cashflows generated from taking the long position in IO/PO synthetics mimicked the cashflows from holding the reference cash bonds themselves.

This synthetic market offered advantages that were previously not present in the cash market. Because settlement of IO/PO synthetics did not necessitate any transfer of bonds, there was significantly greater liquidity in investing in the synthetic markets. Notional balances could in theory exceed the total supply of cash bonds themselves, giving more investors access to this type of exposure. In addition, coupon cashflows in an IOS reflected interest generated across a number of pools under a particular cohort. This means that the long investor needed to only worry about the performance of the pools collectively, instead of having to focus on individual pool characteristics such as prepayment speeds. Lastly, since there was no upfront capital commitment (outside of having margin), returns generated from these synthetic structures were inherently levered.

XVII. (Legacy) Structures that Jump

While the PO market offers portfolios a set of instruments that improve returns if prepayments speed up, not all portfolios can go without a coupon on their investment. Many need cash flow to cover the cost of financing. For these portfolios, the CMO market formerly produced jump classes. These classes, usually structured as one of several support classes in a PAC/support deal, jump to the front of the line for receiving principal if certain prepayment conditions are met. The jump classes usually trade below par, and the buyer profits whenever the class jumps to the front of the line and gets its principal back at par. The art of investing in jump classes revolves around understanding their jump triggers.

Figure 73: Structures that jump essentials

Key Features of Classes that Jump	Payment rules that allow one or more classes to jump over others and become the first in line to receive principal from amortization, prepayments or both The single or cumulative prepayment speeds that trigger the jump class to move to the front of the line The conditions, if any, that require the class to move back from the front of the principal line The underlying collateral Distance of the price from par
Common Uses	If priced below par, to add to portfolio return if prepayments meet or exceed the jump trigger To provide a high coupon if prepayments remain below the jump trigger

Source: J.P. Morgan

Every CMO amounts to a set of rules for allocating principal among outstanding classes, and some CMOs change the order in which classes receive principal based on explicit prepayment rates. If prepayments meet or exceed a predetermined rate, a jump class gets principal before the others. A CMO that included support classes A, B and C, for example, might pay principal in exactly that order unless prepayments exceed a certain rate. Once prepayments hit the trigger, class C might jump to the front of the line and get principal before the others. Class C could either stay at the front of the line until retired or revert to the normal order—a difference that depends on how the jump rules are written.

Jump rules usually apply to Z or accrual classes structured off of support cash flows, an approach where small changes in payment priority can make big differences in average life and performance. The Z feature lowers the dollar price of the jump class and adds to its performance if it jumps to the front of the line and starts returning principal at par. The rules can have a big effect not only on the class that jumps, but also on the classes that consequently get pushed back in the principal line.

Classes that jump fall into two broad categories:

- Sticky jump classes
- Non-sticky jump classes

Sticky jump classes jump to the front of the principal line and stay or stick there until retired, hence the name. The rules that trigger a sticky jump can be either cumulative or noncumulative. A cumulative rule measures prepayments since deal issuance and compares the total against some threshold. If the total exceeds the threshold, the class jumps and keeps getting principal until retired, even if prepayments slow. A noncumulative rule, which is relatively rare, triggers the jump class if prepayments exceed a threshold only for a single reporting period. Once the noncumulative trigger is hit, the jump class moves to the front of the principal line and stays there until retired.

Non-sticky jump classes have potential to revert back to their original position. The rules that trigger a non-sticky jump can also be cumulative or noncumulative. A cumulative trigger will specify a prepayment threshold for each period over the potential life of a security. For example, a deal could include a class that jumps if life prepayment

speeds exceed 300 PSA. The rules compare the prepayments in the underlying collateral to the running threshold and then determine whether the non-sticky jump class moves to the front of the line or not. If cumulative prepayments exceed the threshold, the class jumps. Otherwise, it stays in its original position. A non-sticky jump class might jump one period, but, if prepayments slow enough, revert to its original payment priority either in the next period or over the next several periods.

Some non-sticky jump classes have multiple rules that determine whether they jump or not, and sometimes these rules make the non-sticky class hard to unstick. For example, the rules might require a non-sticky class to jump if cumulative prepayments exceeded either 300 PSA or 6 CPR. Depending on how prepayments evolve over the life of the underlying collateral, either one or both of the rules might trigger the class to jump. Even though the rules technically allow a class to revert back to its original payment priority, a series of rules can make the odds of a jump so high that it effectively becomes a sticky jump class.

Finally, both sticky and non-sticky jump classes can affect the performance of other classes in a deal. If the jump class is a Z bond, then its coupon typically is used to pay down principal in some other class in the deal. That keeps the average life of the receiving class shorter than it would be otherwise. If the jump class moves to the front of the principal line, then not only does the average life of other classes tend to extend, but any classes that would have received the Z bond's coupon extend even more. That can make the average life of jump-Z-affected classes highly variable.

Jump classes can add impressive returns to a portfolio that anticipates either a short-term rise in prepayments or consistently fast speeds. Jump Z classes can work particularly well if priced below par. Since the jump class receives principal back at par, a jump Z that hits its trigger can add materially to portfolio returns in a short amount of time. If the jump class fails to hit its trigger, however, it may remain outstanding for quite a long time. The yield of a jump Z nevertheless can help offset any potential financing costs—something a PO has a harder time doing.

Portfolios with IOs, premium-priced MBS, or with other securities subject to prepayment risk could use jump classes to hedge some of that exposure. The rise in prepayments might hurt the at-risk securities but help the performance of the jump class.

XVIII. (Legacy) Non-Accelerated Senior (NAS) classes

Non-accelerated senior or NAS classes offered a lot of the same features as a PAC, but with one important difference—structure that stands up better to wide swings in prepayments. Although PACs return principal predictably if prepayments stay within a predetermined range, wide swings in speeds can erode the PAC bands and leave an investor holding a cash flow almost identical to a sequential class. For investors trying to match MBS assets against liabilities, eroding PAC bands create challenges. And for investors managing for total return, eroding bands can hurt price performance. NAS classes avoided the challenges of PAC band drift while still offering principal lockout and the flexibility to customize principal windows and average lives.

Figure 74: NAS class essentials

Key Features of NAS Classes	Payment rules that lock out the NAS class from receiving its full share of prepayments, amortization or both
	The size of the NAS class as a percent of the total deal—a signal of the stability of the NAS class' average life
	The underlying collateral
Common Uses	Reduce portfolio prepayment risk
	Match the average life or duration of a liability
	Manage yield curve exposure

Source: J.P. Morgan

A NAS class always gets a percentage of the total principal generated by the underlying collateral, but that percentage changes over time. The rules usually define the NAS percentage as a fraction of the principal that would flow to the class if it simply received its pro-rata or fair share of amortization and prepayments. In a month where the unpaid principal balance in the NAS class amounted to 20% of total unpaid principal in the deal, the pro-rata amount would be 20% of any repaid principal. If the underlying collateral generated \$10 million in amortization and prepayments one month, for instance, the pro-rata amount due to the NAS would be \$2 million. The percentage actually paid to the NAS, however, is usually a fraction of the pro-rata amount. If the underlying collateral generates \$10 million one month, a 0% share of the pro-rata amount would tally to \$0, a 20% share to \$400,000 and a 100% share to \$2 million. Classic NAS receive 0% of the pro-rata amount for the first five years, 30% for the sixth year, 40% for the seventh, 60% for the eighth, 80% for the ninth and 100% for the tenth and later years. This is called a shifting interest schedule since the NAS class's interest in or share of pro-rata principal shifts or changes over time. Of course, in situations where the NAS is the only remaining class in a deal, it gets 100% of any repaid principal regardless of the shifting interest schedule.

By assigning the NAS class a 0% share for its first few years, the rules effectively lock out the class from receiving any principal. During this period, every other class in a deal would have to pay down before the NAS would get its first dollar of principal. That can buffer the NAS from dramatic swings in prepayments. Even as the NAS share rises, as long as the tranche receives less than 100% of its pro-rata amount, the structure dampens the impact of collateral amortization and prepayment on the investor. Of course, the quality of that buffering depends on the percentage of the total structure that lies outside the NAS class. The smaller the NAS class, then the more collateral there is outside the tranche to buffer it against swings in prepayments.

NAS classes served the same investment purposes as PACs, adding stability to the return of portfolio principal. For asset-liability managers, that stability could help the portfolio pay off liabilities scheduled for certain periods or can help avoid the reinvestment risk that would come with uncertain principal return. For managers focused on total returns, the stability of NAS principal made the structure useful for creating exposure to different parts of the yield curve. Of course, since NAS offered more reliable principal repayment than pass-throughs or even PACs, they often carried a lower yield in return.

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